FINANCE 656: Global Asset Allocation

# Follow the Science

Campbell R. Harvey
Duke University and NBER

# Setting

Researchers like low “p-values” (probability value)

- In hypothesis testing, if an effect is “two sigma” (e.g., from zero) this is associated with a p-value of 0.05 or 5%. You can think of the two sigma as a t-statistic>2.
- Ideally, we like to see p<0.05 to declare something “significant”
- However, there many reasons not to “follow the science”.






Source: [https://xkcd.com/882/](https://xkcd.com/882/)




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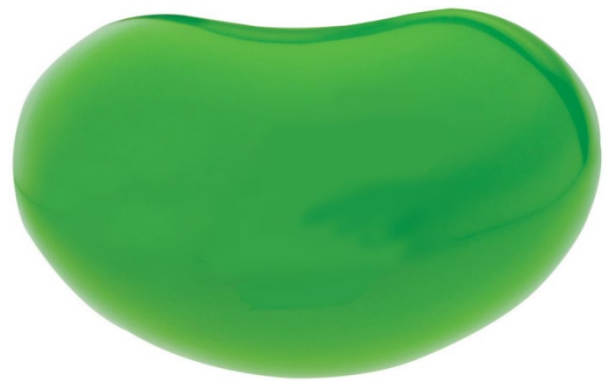




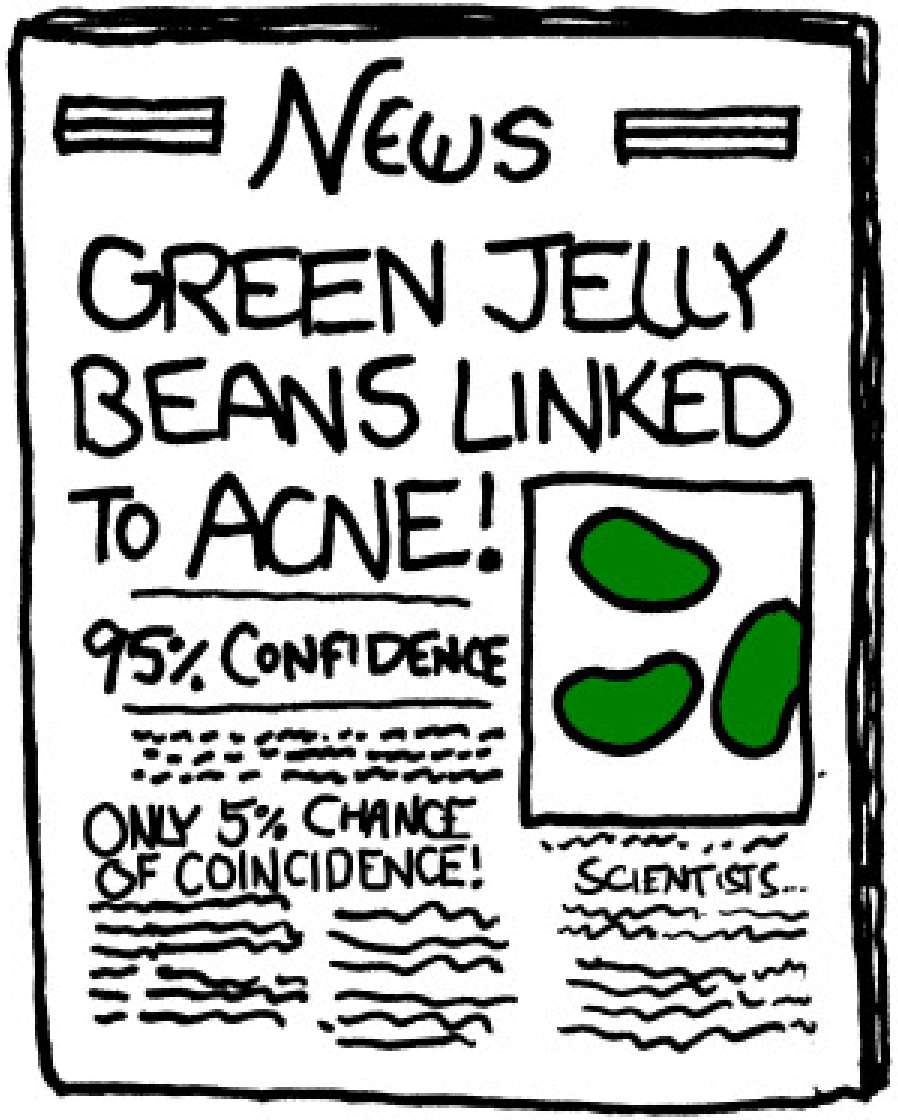

Source: [https://xkcd.com/882/](https://xkcd.com/882/)

Skip 17 panels of more negative tests, all p-values>0.05



Source: [https://xkcd.com/882/](https://xkcd.com/882/)



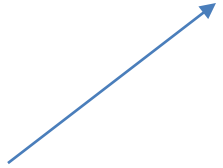

Source: https://xkcd.com/882/

# Examples in Financial Economics

Two sigma rule only appropriate for a single test

- As we do more tests, there is a chance we find something "significant" (by the two sigma rule) but it is a fluke.
- Here is a simple way to see the impact of multiple tests for a two sigma test:

| # of tests | 1 | 5 | 10 | 20 | 26 | 50 | n |
|---|---|---|---|---|---|---|---|
| Prob of fluke | 5% | 23% | 40% | 64% | 74% | 92% | 1-0.95^n |



XKCD Jelly Beans and Acne

# Examples in Financial Economics

The promotional email:

- You get an email at the end of each month from an investment manager with “judge my record” as a slogan
- The email recommends either a long or a short position in the S&P
- After receiving 10 correct recommendations in a row, you switch your investment account to the new manager

# Examples in Financial Economics

The promotional email

- Later out, you find out (the hard way) the promotional strategy
- Manager sends out each month 100,000 emails:

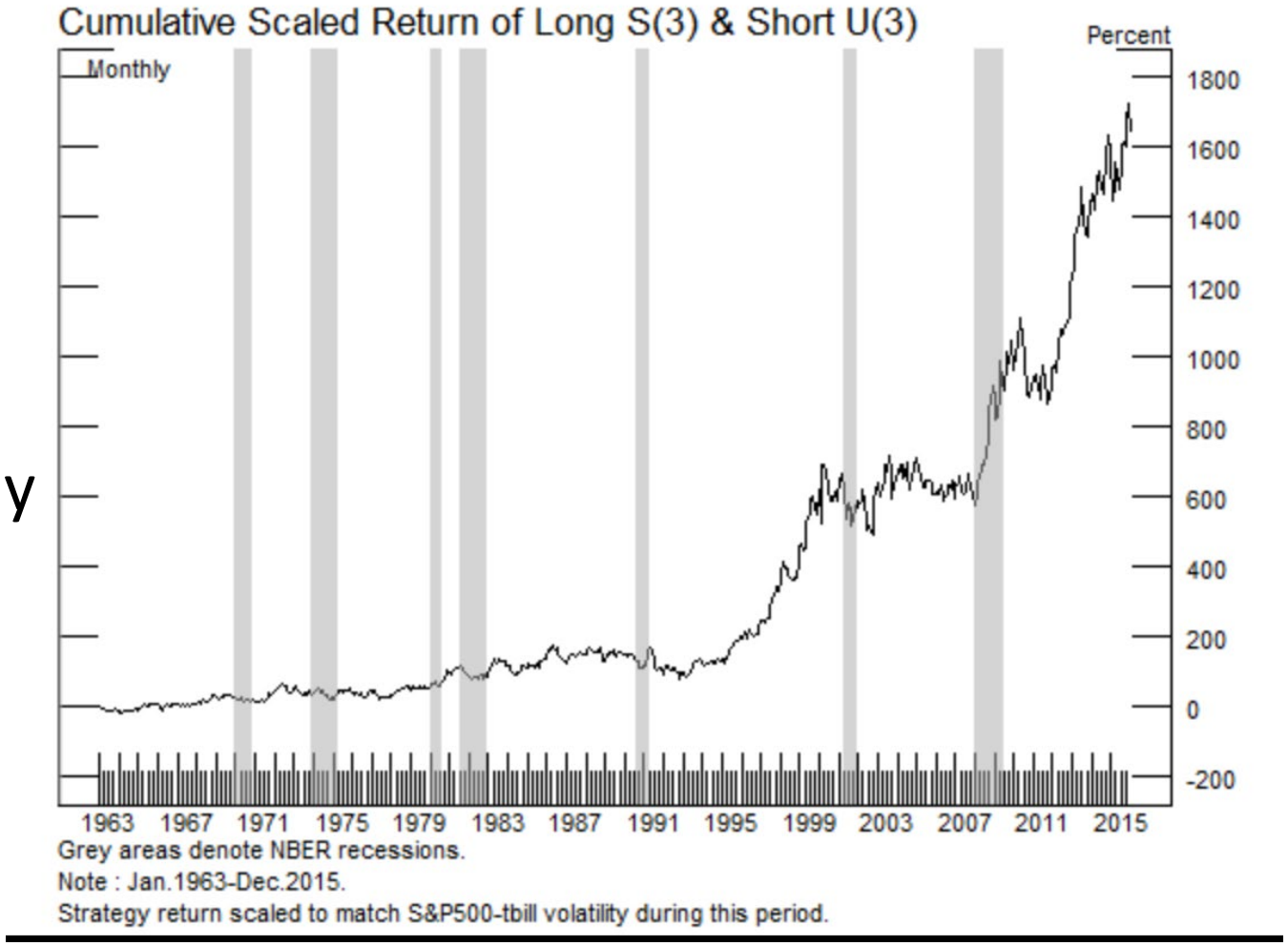
  **50,000 saying long and 50,000 short**

- The next month manager sends only to those who got the correct prediction, so next month is 25,000 long and 25,000 short recommendations
- 97 people will get 10 correct in a row ($100{,}000 \times 0.5^{10}$ )
- No skill here. It is random.

# Examples in Financial Economics

## 3.4 sigma strategy

- Profitable during fin crisis
- Zero beta vs. market, value, size, and momentum
- Impressive performance recently



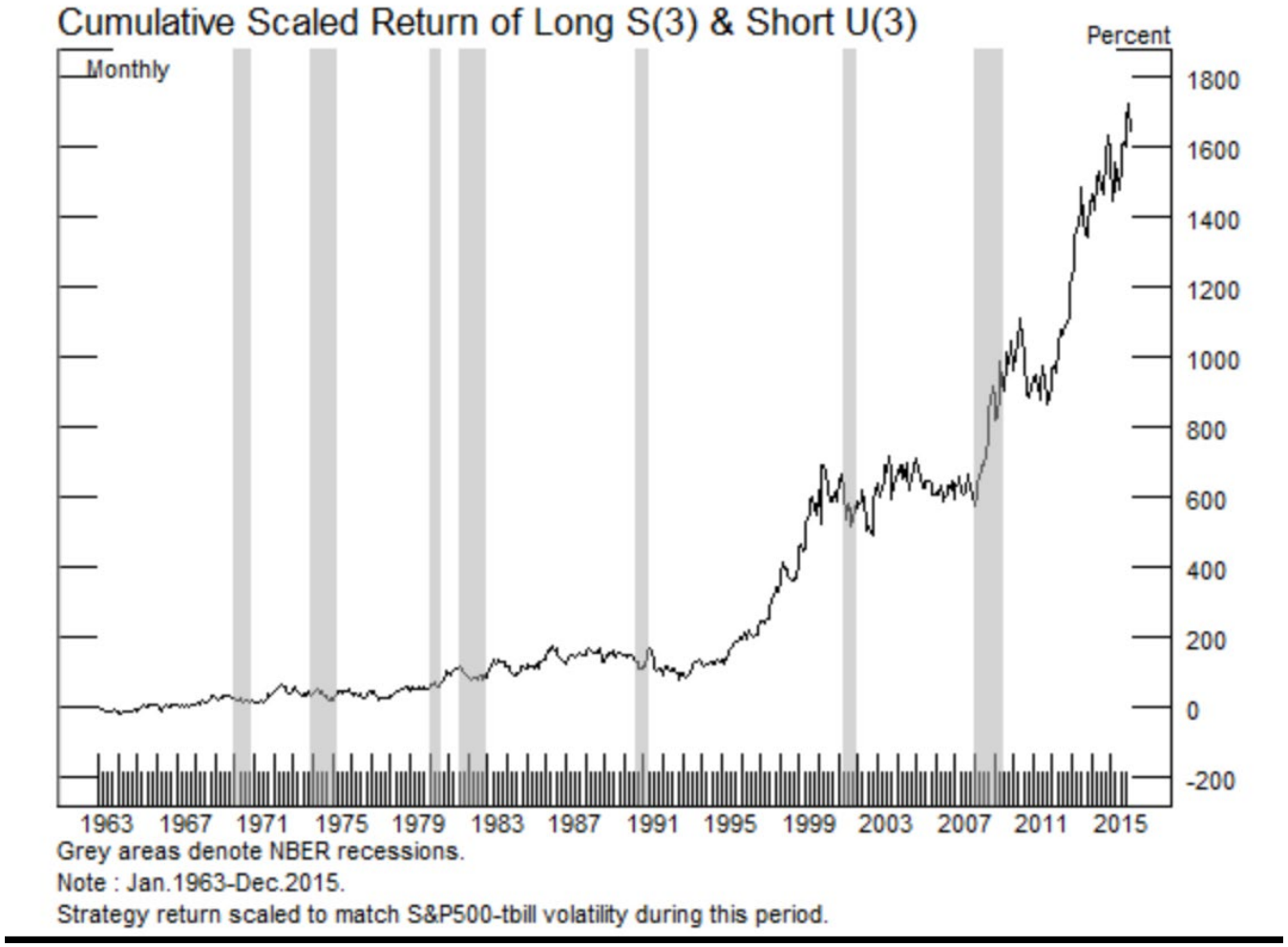



Campbell R. Harvey, "The Scientific Outlook in Financial Economics", Presidential Address, American Finance Association, 2017

# Examples in Financial Economics

## Details

- Long tickers “S”
- Short tickers “U”



# Examples in Financial Economics

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XKCD Jelly Beans and Acne

Alphabet, i.e., ticker symbols

# Examples in Financial Economics

Research

- Companies with meaningful ticker symbols, like Southwest's LUV, and show they outperform.[1]
- There is another study that argues that tickers that are easy to pronounce, like BAL vs. BDL, outperform in IPOs.[2]
- There is yet another study that suggests that tickers that are congruent with the company's name, outperform.[3]



[1] Head, Smith and Watson, 2009; [2] Alter and Oppenheimer, 2006; [3] Srinivasan and Umashankar

# Examples in Financial Economics

## 5 factors

# A five-factor asset pricing model ☆

Eugene F. Fama [a], Kenneth R. French [b,*]

[a] *Booth School of Business, University of Chicago, USA*
[b] *Tuck School of Business, Dartmouth College, Hanover, NH 03750, USA*

Journal of Financial Economics 116 (2015) 1–22




A B S T R A C T

A five-factor model directed at capturing the size, value, profitability, and investment patterns in average stock returns performs better than the three-factor model of Fama and French (FF, 1993). The five-factor model's main problem is its failure to capture the low

# Examples in Financial Economics

## 15 factors

*Critical Finance Review, 2015, 4: 1–44*

**The Cross-section of Expected Stock Returns**

Jonathan Lewellen, Dartmouth College and NBER

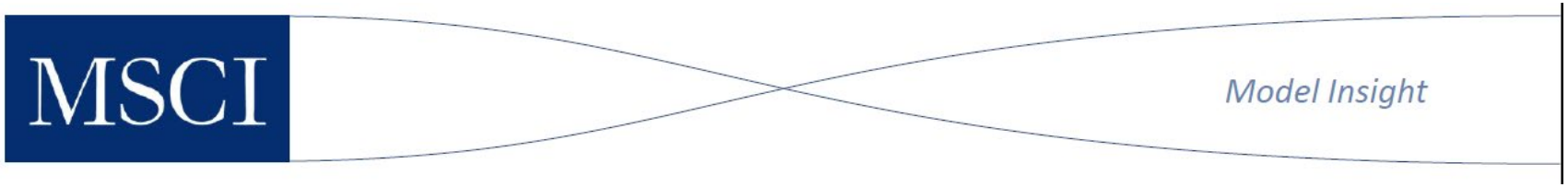



*Model 3: Fifteen predictors*

| | | |
|---|---|---|
| $LogSize_{-1}$ | −0.73 | −3.67 |
| $LogB/M_{-1}$ | 0.98 | 2.92 |
| $Return_{-2,-12}$ | 3.46 | 3.99 |
| $LogIssues_{-1,-36}$ | −2.85 | −4.30 |
| $Accruals_{Yr-1}$ | −6.12 | −4.56 |
| $ROA_{Yr-1}$ | 6.56 | 2.95 |
| $LogAG_{Yr-1}$ | −3.45 | −5.13 |
| $DY_{-1,-12}$ | −25.04 | −2.55 |
| $LogReturn_{-13,-36}$ | −0.77 | −1.78 |
| $LogIssues_{-1,-12}$ | −1.41 | −1.07 |
| $Beta_{-1,-36}$ | 1.57 | 2.39 |
| $StdDev_{-1,-12}$ | −9.47 | −1.67 |
| $Turnover_{-1,-12}$ | −22.12 | −3.49 |
| $Debt/Price_{Yr-1}$ | −0.29 | −1.48 |
| $Sales/Price_{Yr-1}$ | 0.13 | 1.58 |
| N | 2,964 | |

# Examples in Financial Economics

## 82 factors




### Non-linear Size

Definition: *1.0 · NLSIZE*

*NLSIZE* *Cube of Size*

First, the standardized Size exposure (i.e., log of market cap) is cubed. The resulting factor is then orthogonalized to the Size factor on a regression-weighted basis. Finally, the factor is winsorized and standardized.

### Non-linear Beta

Definition: *1.0 · NLBETA*

*NLBETA* *Cube of Beta*

First, the standardized Beta exposure is cubed. The resulting factor is then orthogonalized to the Beta factor on a regression-weighted basis. Finally, the factor is winsorized and standardized.

Source: The Barra US Equity Model (USE4), MSCI (2014)

# Examples in Financial Economics

## 400 factors!




Maximize Your Analytical Efficiency and Insight

The Alpha Factor Library is the industry's most comprehensive source of value-added stock signals and multi-factor stock-selection models. Available as a data feed and accessible through the S&P Capital IQ platform, the factor library contains over 400 signals that can be used to jump start new investment products or dramatically improve existing ones.

**Key Advantages:**

- Access over 450 quantitative stock selection signals spanning seminal academic literature and the latest practitioner expertise

Enabling the High-Performing Quantitative Investor

Source: https://www.capitaliq.com/home/who-we-help/investment-management/quantitative-investors.aspx

# Examples in Financial Economics

## 18,000 signals examined in Yan and Zheng (2015)

We construct a universe of fundamental signals by imitating the search process of a data snooper. We start with all accounting variables in Compustat that have a sufficient amount of data. We then use permutational arguments to construct over 18,000 fundamental signals. We choose

# Examples in Financial Economics

**2.1 million!**

## $p$-hacking: Evidence from two million trading strategies

Tarun Chordia  Amit Goyal  Alessio Saretto*

August 2017

**Abstract**

We implement a data mining approach to generate about 2.1 million trading strategies. This large set of strategies serves as a laboratory to evaluate the seriousness of

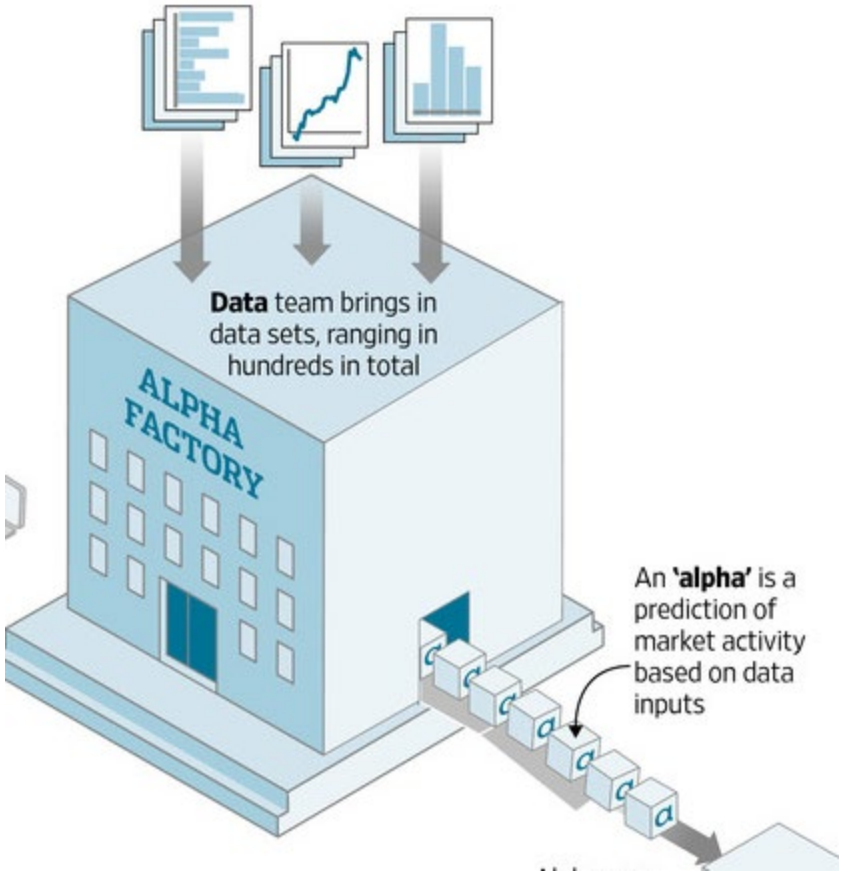
# Examples in Financial Economics

THE WALL STREET JOURNAL.

U.S. Edition ▾ | April 13, 2017 | Today's Paper

**100 million!?**

"…the company has four million "alphas" to date and is aiming for 100 million."



# What's going on?

Forces causing mistakes

1. Failure to account for luck + evolutionary propensity not to account for luck
2. Failure in specifying and conducting scientific tests
3. Failure to take rare effects into account

# A framework to separate luck from skill

*Six research initiatives*:*

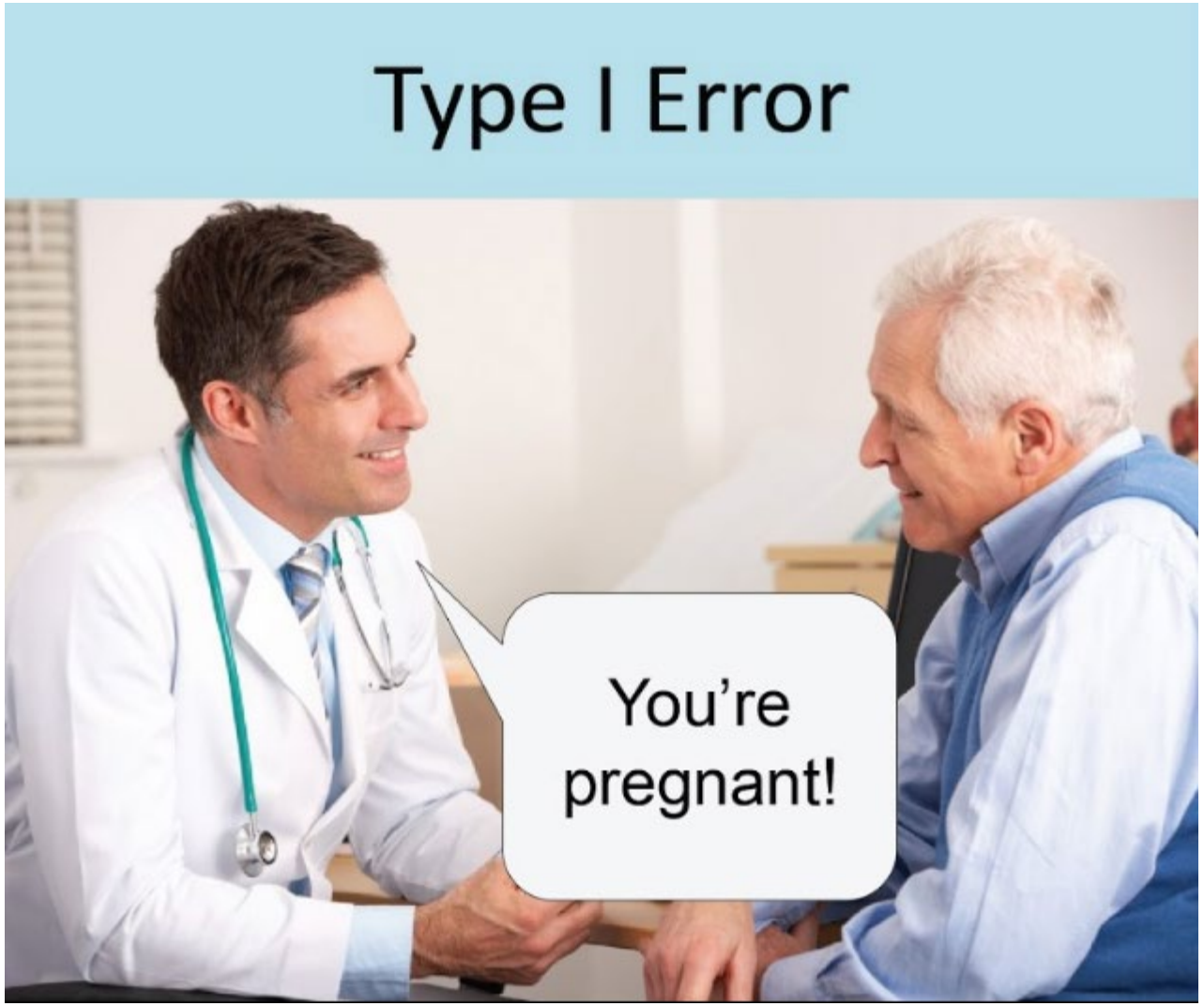
1. Explicitly adjust for multiple tests ("…") [**RFS** 2016]
2. Bootstrap ("Lucky Factors") [**JFE**, 2021]
3. Noise reduction ("Detecting Repeatable Performance") [**RFS** 2018]
4. Rare effects ("Presidential Address") [**JF** 2017]
5. Idiosyncratic noise ("Alpha dispersion") [**JFE** 2019]
6. **"False (and missed) discoveries" [JF 2020]**

*Bibliography on last page. All my research at: https://papers.ssrn.com/sol3/cf_dev/AbsByAuth.cfm?per_id=16198

# Luck

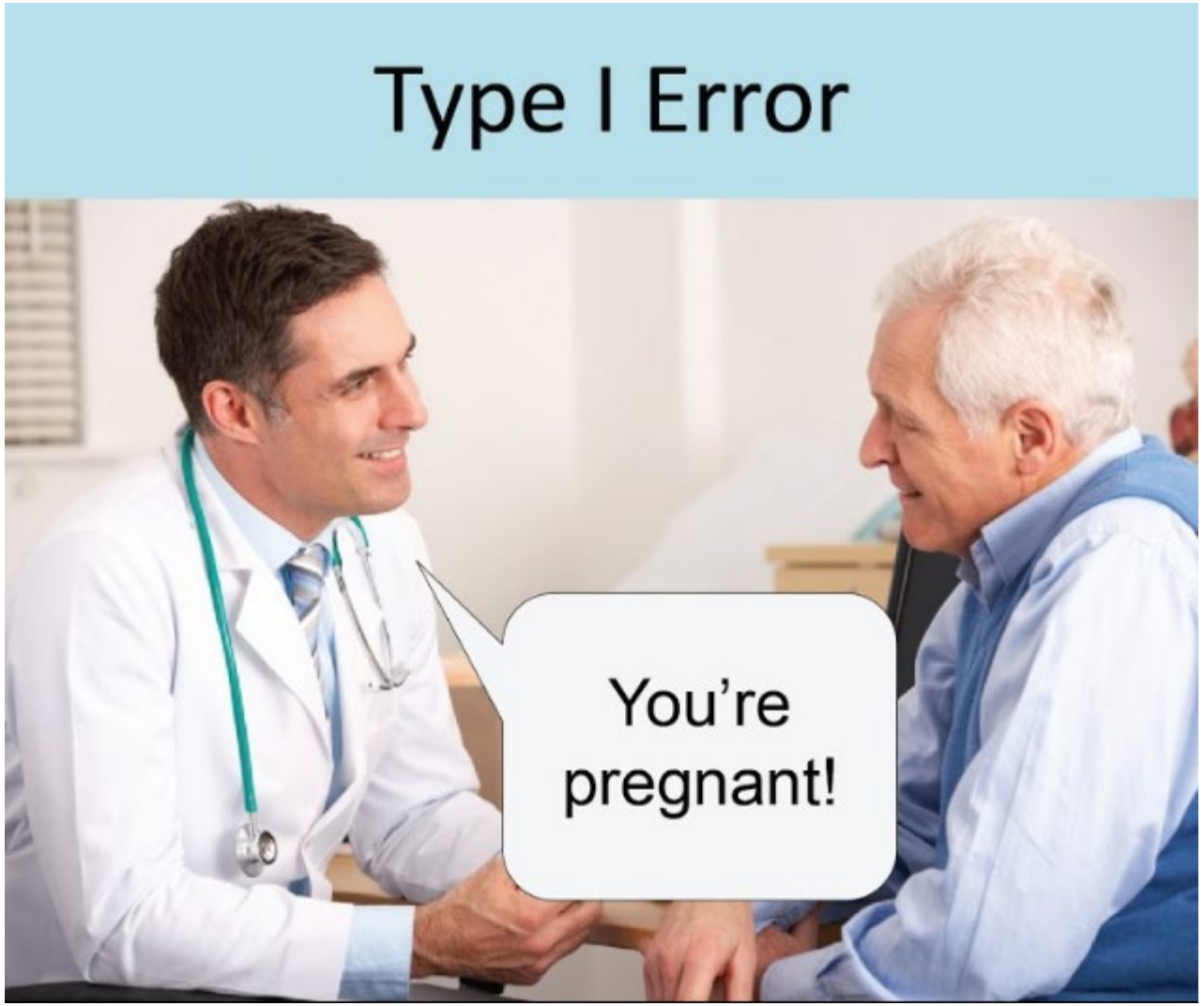
- Why are we so easily fooled by randomness?

# Terminology



# Terminology

I thought this manager was skilled but that was a mistake: **False Positive**



# Terminology



# Terminology




I didn't invest in this manager but that was a mistake **False Negative**

Type II linked to Type I.

- For example, if all patients declared pregnant, there is no Type II error.

# Evolutionary Foundations



Rustling sound in the grass ….

# Evolutionary Foundations





Type I error

Rustling sound in the grass ….

# Evolutionary Foundations



Type II error

# Evolutionary Foundations



Type II error

In examples, cost of Type II error is large – potentially death.

# Evolutionary Foundations

- High Type I error (low Type II error) animals survive
- This preference is passed on to the next generation
- This is the case for an evolutionary predisposition for allowing high Type I errors

# Evolutionary Foundations




B.F. Skinner 1947

Pigeons put in cage. Food delivered at regular intervals – feeding time has nothing to do with behavior of birds.

# Evolutionary Foundations

## Results

- Skinner found that birds associated their behavior with food delivery
- One bird would turn counter-clockwise
- Another bird would tilt its head back

# Evolutionary Foundations

## Results

- A good example of overfitting – you think there is pattern but there isn't
- Skinner's paper called:
  - 'Superstition' in the Pigeon, JEP (1947)
- But this applies not just to pigeons or gazelles…

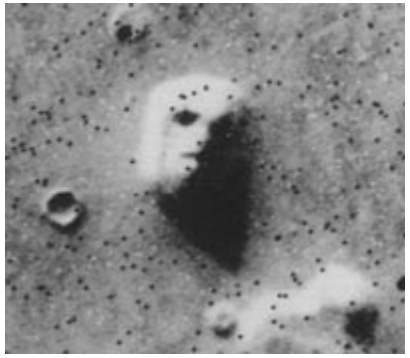
# Evolutionary Foundations



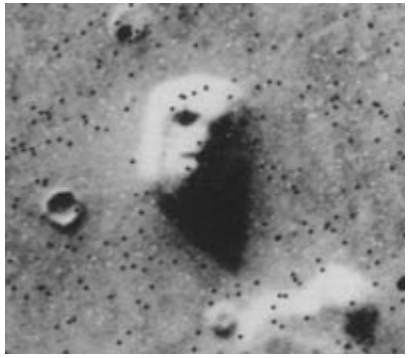
Klaus Conrad 1958

Coins the term Apophänie. This is where you see a pattern and make an incorrect inference. He associated this with psychosis and schizophrenia.

# Evolutionary Foundations

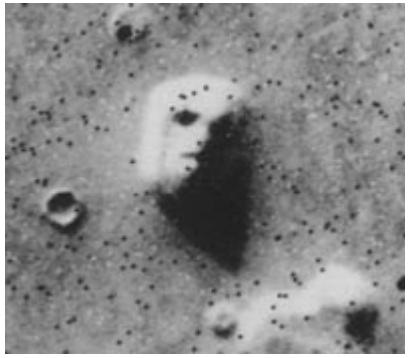


# Evolutionary Foundations





# Evolutionary Foundations







# Evolutionary Foundations

- Apophany is a Type I error (i.e., false insight)
- Epiphany is the opposite (i.e., true insight)
  - Apophany may be interpreted as overfitting

*"....nothing is so alien to the human mind as the idea of randomness." --John Cohen*

K. Conrad, 1958. *Die beginnende Schizophrenie. Versuch einer Gestaltanalyse des Wahns*

# Evolutionary Foundations

- Sagan (1995):
  - As soon as the infant can see, it recognizes faces, and we now know that this skill is hardwired in our brains.



C. Sagan, 1995. *The Demon-Haunted World*

# Evolutionary Foundations

- Sagan (1995):
  - Those infants who a million years ago were unable to recognize a face smiled back less, were less likely to win the hearts of their parents and less likely to prosper.



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# Evolutionary Foundations

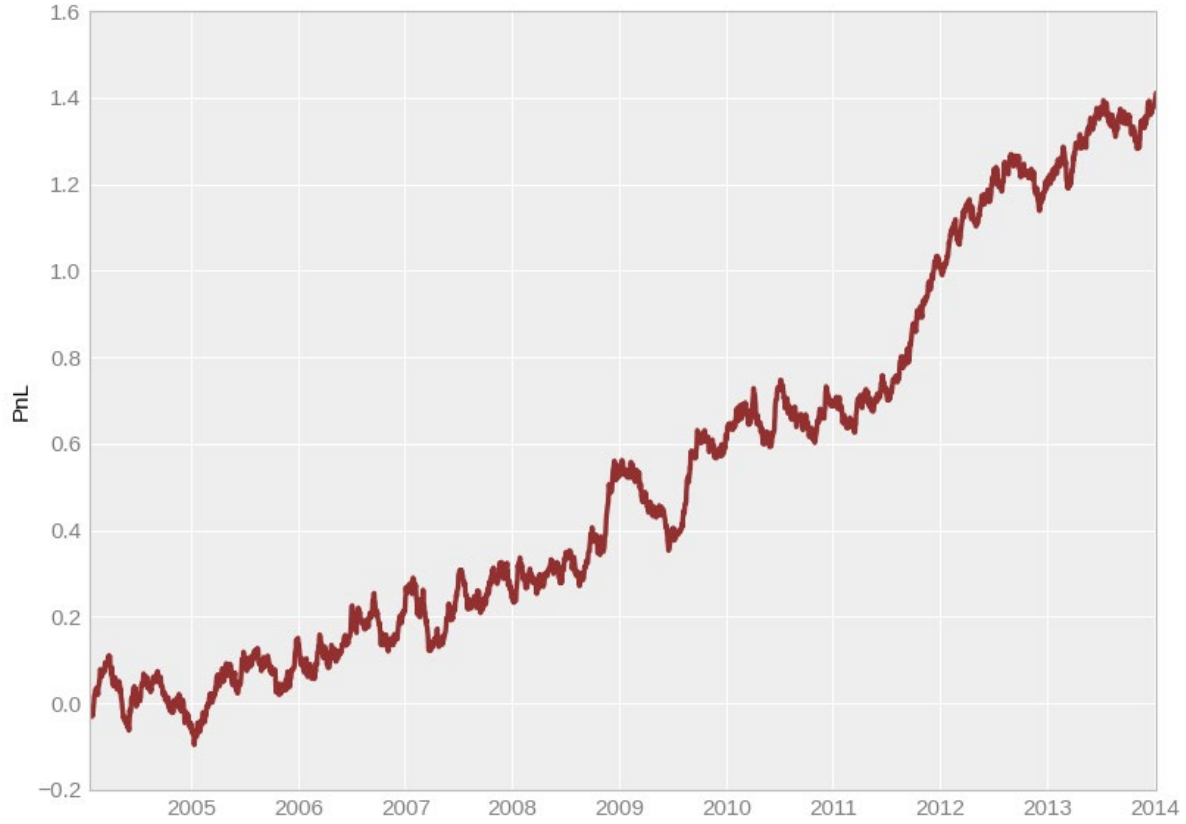
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# What about Finance?

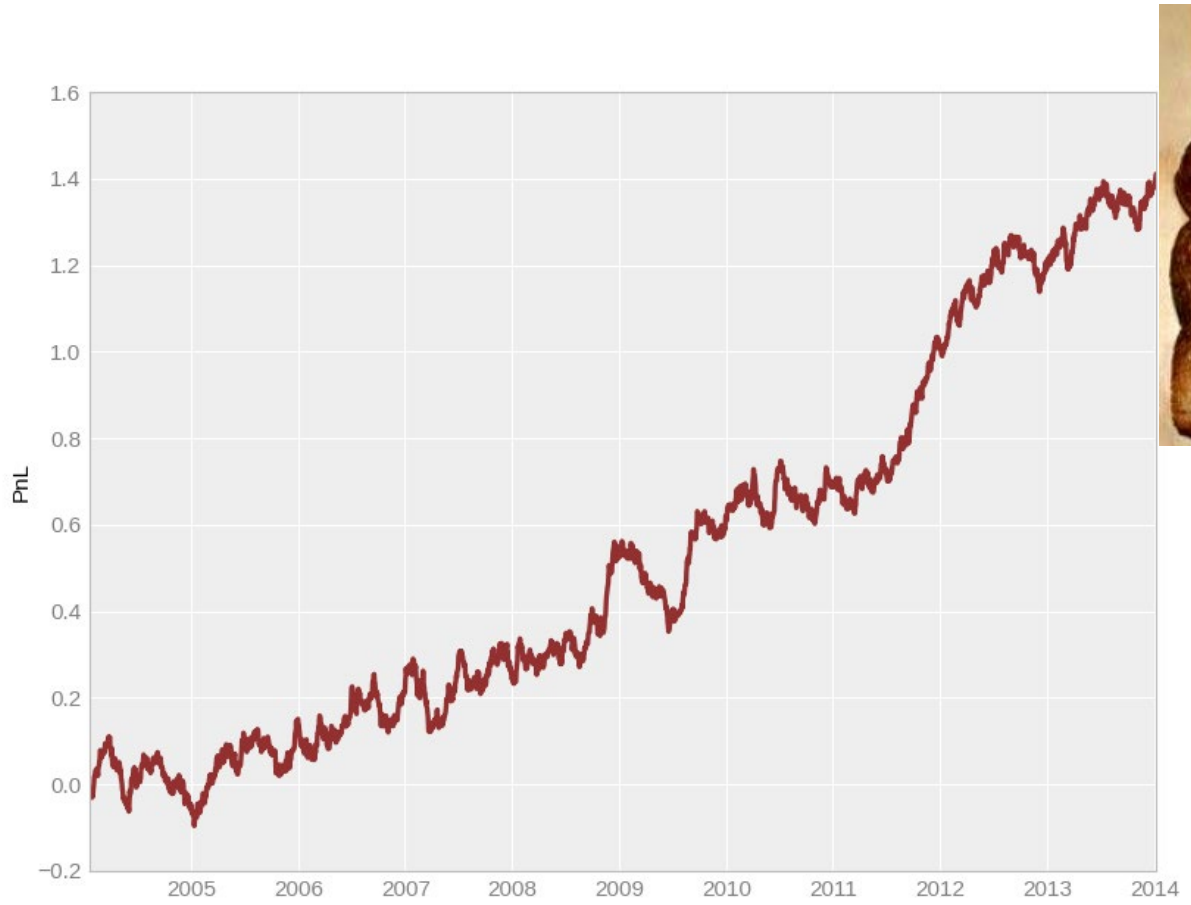



Source: Man-AHL Research

Performance of trading strategy is very impressive.

- SR=1
- Consistent
- Drawdowns acceptable

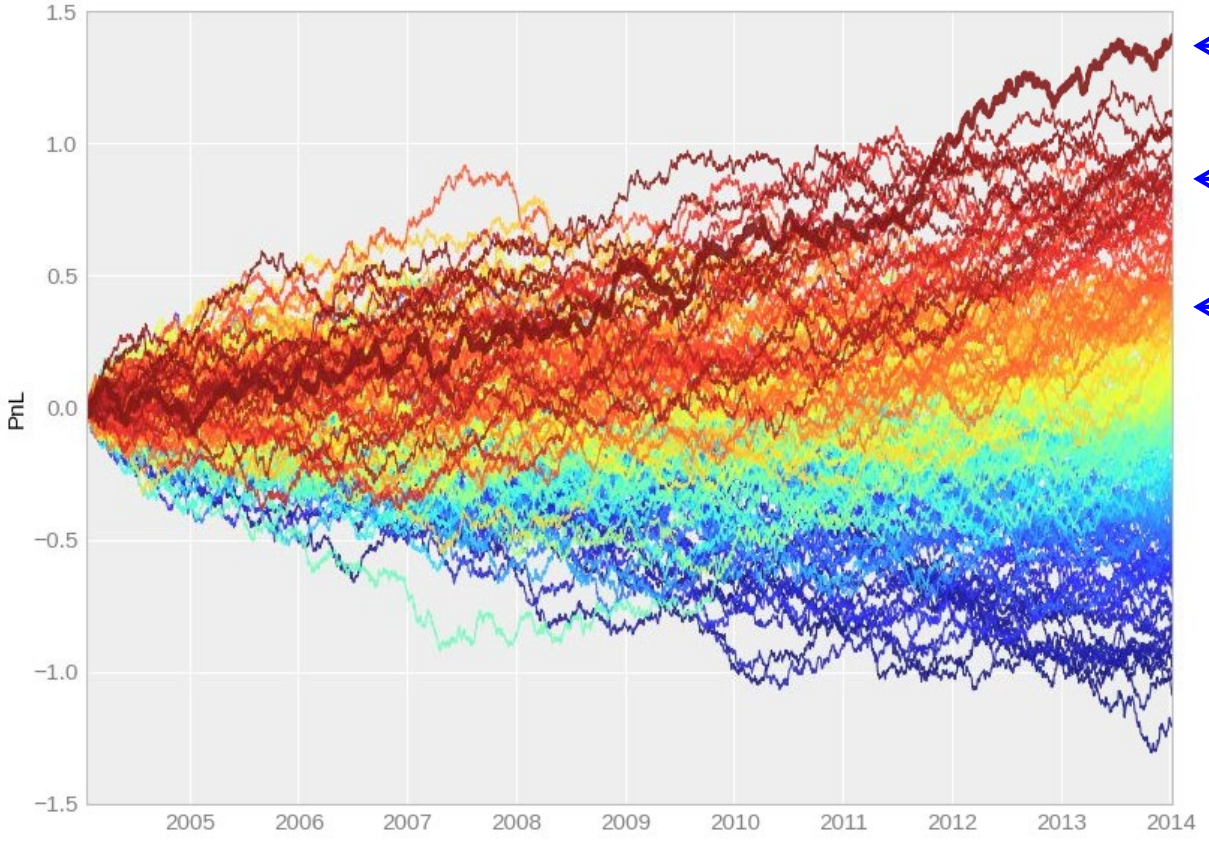
# What about Finance?






Source: Man-AHL Research

# What about Finance?




Sharpe = 1

Sharpe = 2/3

Sharpe = 1/3

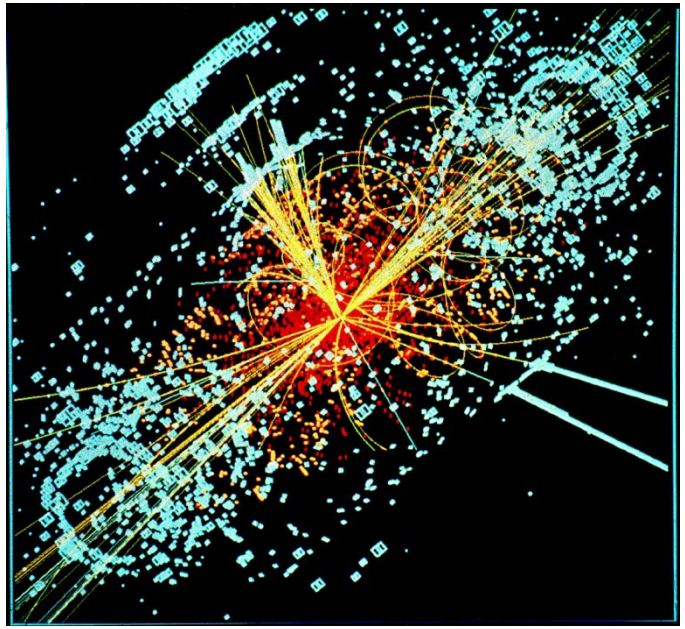
200 random time-series
mean=0; volatility=15%

Source: Man-AHL Research

# Other Sciences?

## Particle Physics

- Higgs' boson proposed in 1964 (same year as Sharpe published the CAPM)
- First tests of the CAPM in 1972
  - Nobel award in 1990.
- Longer road for Higgs:
  - $5 billion to construct LHC.
  - "Discovered" in 2012.
  - Nobel 2013.



# Other Sciences?

## Particle Physics

- Testing method very important
- Particle rare and decays quickly and the key is measuring the decay signature
- Frequency is 1 in 10 billion collisions and over a quadrillion collisions were conducted
- Problem is that the decay signature could also be caused by normal events from known processes

# Other Sciences?

## Particle Physics

- The two groups involved in testing (CMS and ATLAS) decided on what appears to be a tough standard: t-statistic must exceed 5 (i.e., 5-sigma)

# Other Sciences?

## Genetic-Wide Association Studies

- Genetic association research plagued by multiple testing
- Researchers try to link certain diseases to certain genes
- More than 20,000 human genes
- In addition, there is a massive number of combinations of genes
- For the first 10 years of publication of association studies, 98% of published results have been found to be false
  - John Ioannidis: "…*There are millions of scientists, some of whom run millions of analyses in each study they conduct. To avoid false-positive results in genetics, the current goal for a p value should be less than 0.00000005*."*

*Approximately 5.3 sigma

# Other Sciences?

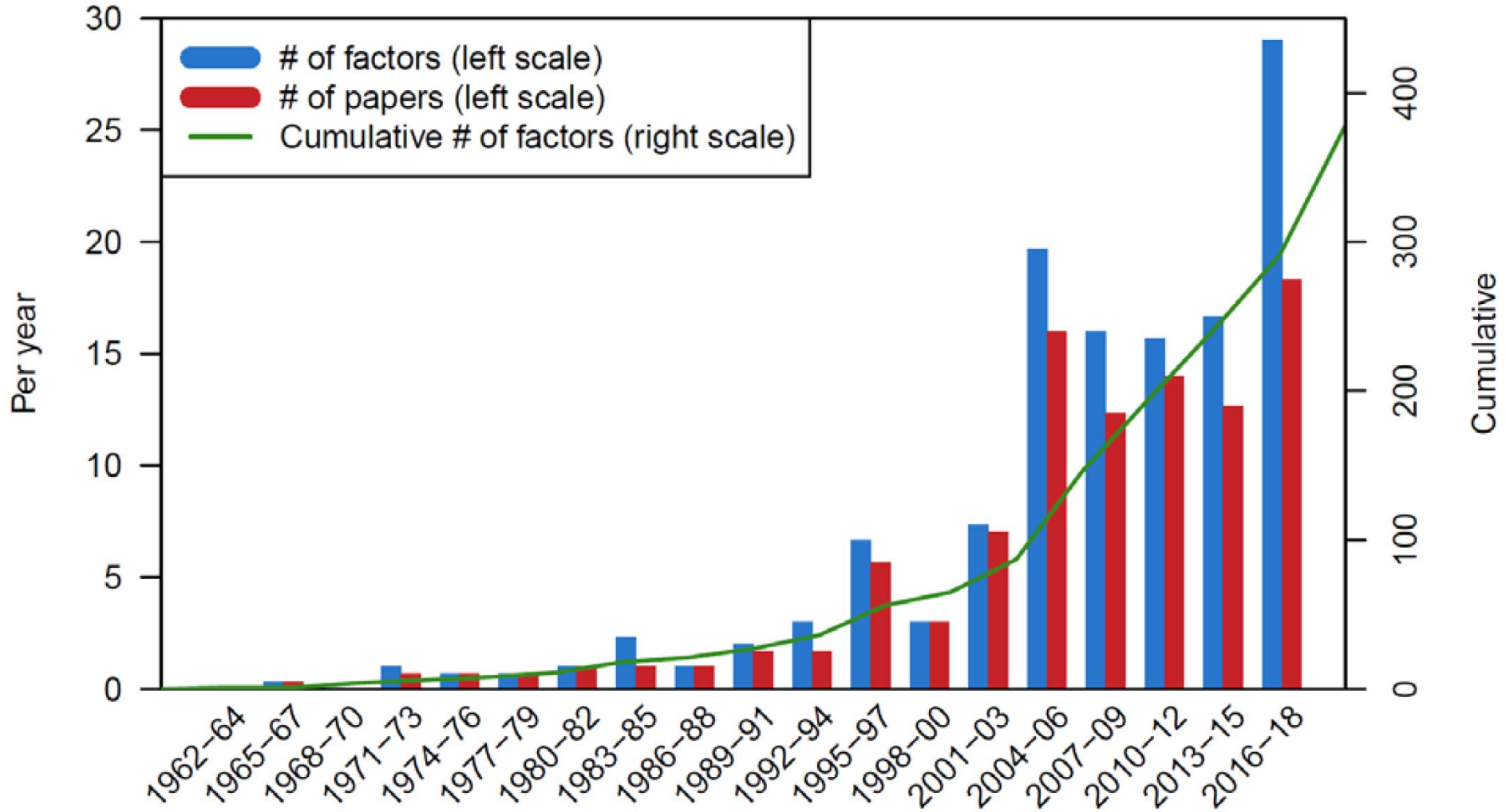
## Genetic-Wide Association Studies

- Recent paper in *Nature* claims two genetic linkages to Parkinson's Disease
- Over 500,000 genetic sequences are tried
- By chance, thousands of sequences will appear to be linked to the disease
- Identified loci had t-statistics>5.3

# 1. Multiple Tests

- Provide a new framework to do multiple tests in the presence of correlations among tests and publication bias (hidden tests)
- Provide guidelines for future research

# 1. Multiple Tests: Number of Factors and Publications




* Journals published through December 2018. Data collection in January 2019.

# 1. Multiple Tests: How Many Discoveries Are False?

- In multiple testing, how many tests are likely to be false?
  - In single testing (significance level = 5%), 5% is the “error rate” (false discoveries)
  - In multiple testing, the false discovery rate (FDR) is usually much larger than 5%

# 1. Multiple Tests: Bonferroni's Method

- Here is a simple adjustment called the Bonferroni adjustment
- For a single test, you are tolerant of 5% false discoveries
- Hence, a p-value of 5% or less means you declare a finding “true”
- Bonferroni simply multiplies the p-value by the number of tests

# 1. Multiple Tests: Bonferroni's Method

- Bonferroni simply multiplies the p-value by the number of tests
- In a single test, if you get a p-value of 0.05 you declare “significant”
- Returning to the Jelly Bean, suppose the green jelly bean test has a p-value of 0.04 – which appears “significant”
- Bonferroni adjustment 20x0.04 = 0.80 which is “not significant” – not even close!

# 1. Multiple Tests: Bonferroni's Method

- Stock market does better under Democratic presidents
- Difference “significant” p-value=.03

**DJIA Returns Under US Presidents Since 1900**

| | | | DJIA Percent Change (%) | |
|---|---|---|---|---|
| President | Start | End | Change as President | Annualized Return |
| T Roosevelt | 9/14/1901 | 3/4/1909 | 21.6 | 2.7 |
| Taft | 3/4/1909 | 3/4/1913 | -1.3 | -0.3 |
| Wilson | 3/4/1913 | 3/4/1921 | -6.9 | -0.9 |
| Harding | 3/4/1921 | 8/2/1923 | 17.4 | 6.9 |
| Coolidge | 8/2/1923 | 3/4/1929 | 255.9 | 25.5 |
| Hoover | 3/4/1929 | 3/4/1933 | -82.8 | -35.6 |
| FDR | 3/4/1933 | 4/12/1945 | 194.4 | 9.3 |
| Truman | 4/12/1945 | 1/20/1953 | 81.7 | 8.0 |
| Eisenhower | 1/20/1953 | 1/20/1961 | 120.3 | 10.4 |
| JFK | 1/20/1961 | 11/22/1963 | 12.2 | 4.1 |
| Johnson | 11/22/1963 | 1/20/1969 | 30.9 | 5.3 |
| Nixon | 1/20/1969 | 8/9/1974 | -16.5 | -3.2 |
| Ford | 8/9/1974 | 1/20/1977 | 23.4 | 8.9 |
| Carter | 1/20/1977 | 1/20/1981 | -0.9 | -0.2 |
| Reagan | 1/20/1981 | 1/20/1989 | 135.1 | 11.3 |
| Bush I | 1/20/1989 | 1/20/1993 | 45.0 | 9.7 |
| Clinton | 1/20/1993 | 1/20/2001 | 226.6 | 15.9 |
| Bush II | 1/20/2001 | 1/20/2009 | -24.9 | -3.5 |
| Obama | 1/20/2009 | 12/31/2014 | 124.1 | 14.5 |
| **Average** | | | **60.8** | **4.7** |
| **Average Republican** | | | **44.8** | **3.0** |
| **Average Democratic** | | | **82.7** | **7.0** |

# 1. Multiple Tests: Bonferroni's Method

- However many possible choices for the test -- here are some:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |
| President | + | House | + | Senate | vs. | President | + | House | + | Senate |

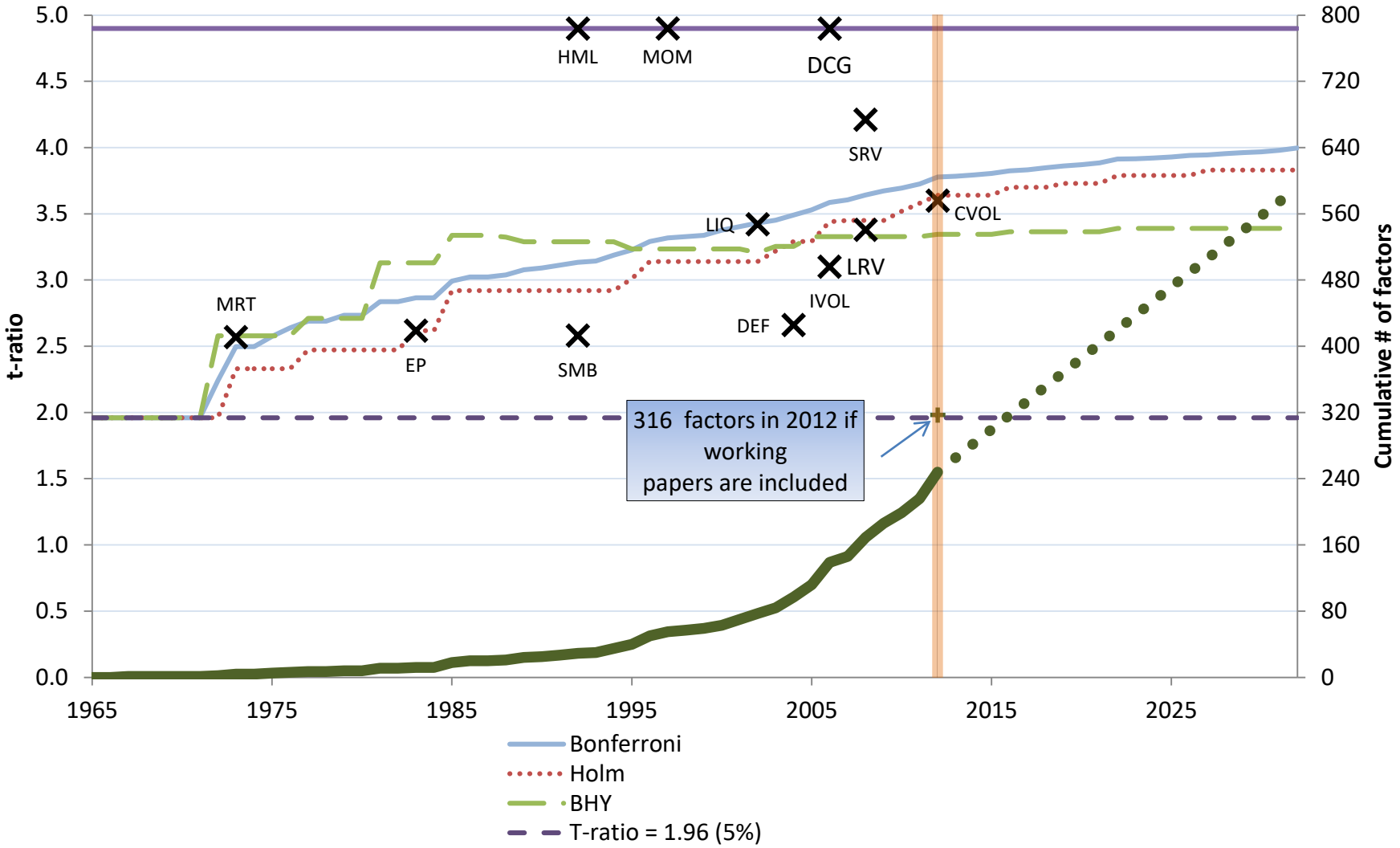
- Bonferroni adjustment eliminates "significant" difference

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| Truman | 4/12/1945 | 1/20/1953 | 81.7 | 8.0 |
| Eisenhower | 1/20/1953 | 1/20/1961 | 120.3 | 10.4 |
| JFK | 1/20/1961 | 11/22/1963 | 12.2 | 4.1 |
| Johnson | 11/22/1963 | 1/20/1969 | 30.9 | 5.3 |
| Nixon | 1/20/1969 | 8/9/1974 | -16.5 | -3.2 |
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Cocquemas and Whaley, 2016

# 1. Multiple Tests: Rewriting History

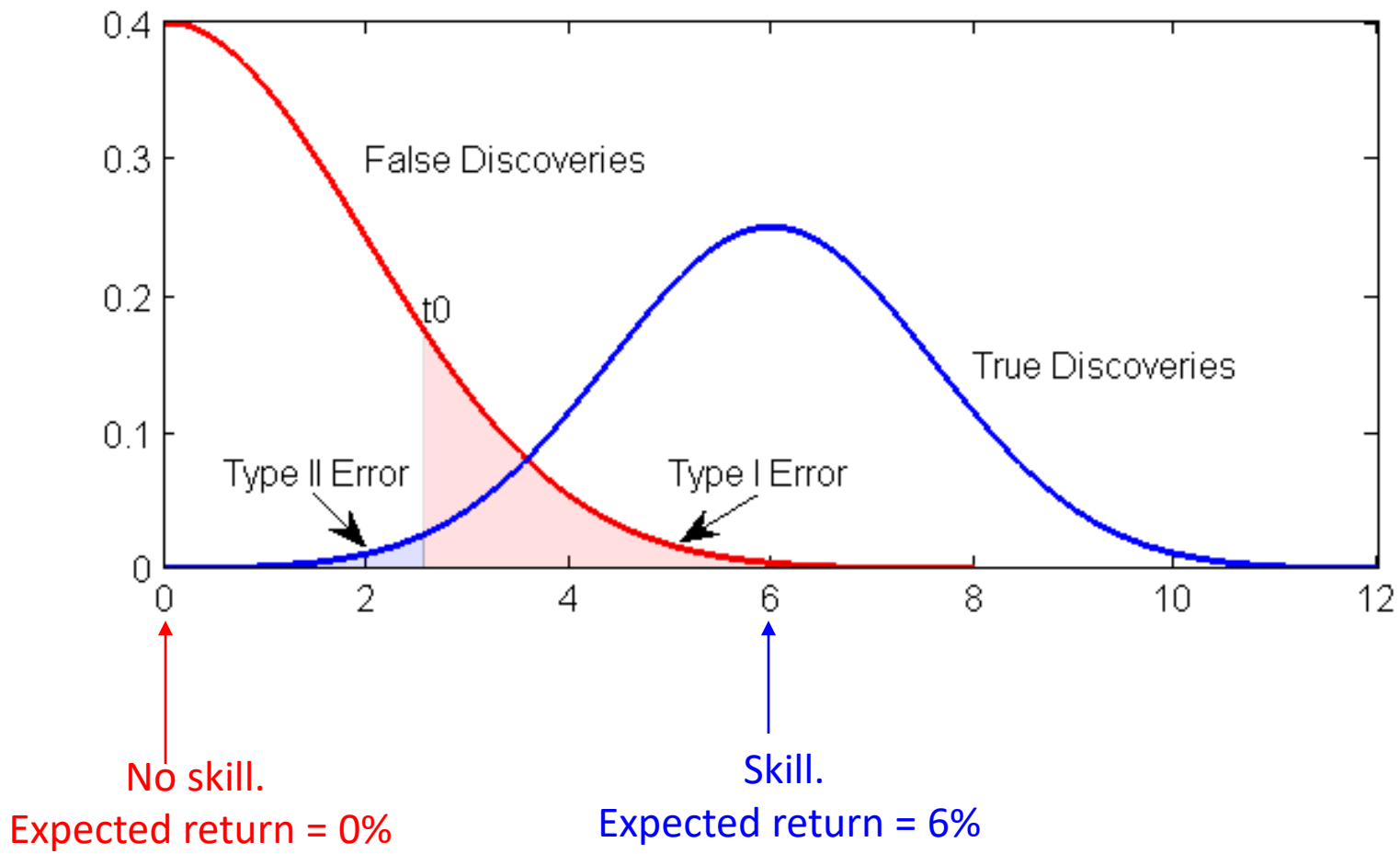


# 1. Multiple Tests: Discussion

However:

- Independence among test statistics is still not dealt with.
- The number of hidden tests seems too low.

# 1. Multiple Tests: A New Framework



# 1. Multiple Tests: Harvey, Liu and Zhu Approach

- Allows for correlation among strategy returns
- Allows for missing tests
- *Review of Financial Studies, 2016*



# 1. Multiple Tests: Backtesting

- Due to data mining, a common practice in evaluating backtests of trading strategies is to discount Sharpe ratios by 50%
- The 50% haircut is only a rule of thumb; we develop an analytical way to determine the haircut

# 1. Multiple Tests: Backtesting

## Method

- Suppose we observe a strategy with an attractive Sharpe Ratio.
- This Sharpe Ratio directly implies a p-value (which roughly tells you the probability that your strategy is a fluke)
- Suppose the p-value is 0.01 which looks pretty good.

# 1. Multiple Tests: Backtesting

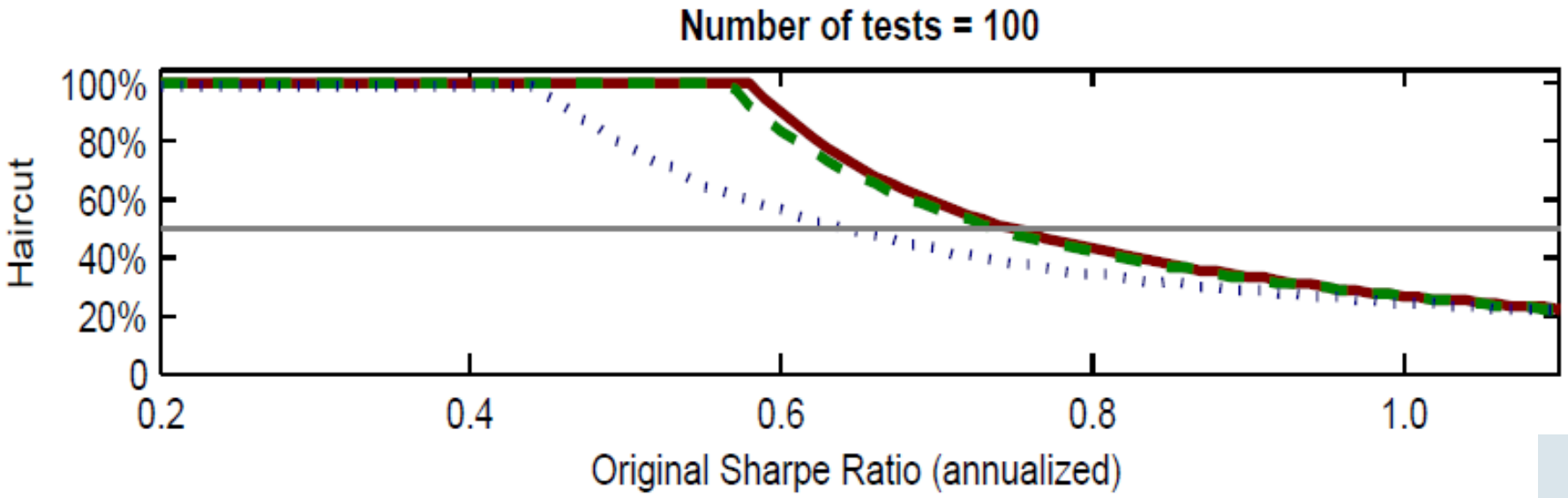
## Method

- However, suppose you tried 10 strategies and picked the best one
- The Bonferroni adjusted p-value is 10x0.01 = 0.10 which would not be deemed “significant”
- Reverse engineer the 0.10 back to the “haircut” Sharpe Ratio*

*Note $\text{Tstat} = \text{SR}\sqrt{\text{T}}$

# 1. Multiple Tests: Backtesting

## Results: Percentage Haircut is Non-Linear




*Journal of Portfolio Management*



# 2. Bootstrapping

Multiple testing approach has drawbacks

- Need to know the number of tests
- Need to know the correlation among the tests
- With similar sample sizes, this approach does not impact the ordering of performance

# 2. Bootstrapping: Lucky Factors

Suppose we have 100 possible fund returns and 500 observations.

- Step 1. Strip out the alpha from all fund returns (e.g. regress on benchmark and use residuals). This means alpha and t-stat exactly equal zero – we have enforced "no skill".
- Step 2. Bootstrap rows of the data to produce a new sheet 500x100* (note some rows sampled more than once and some not sampled at all)

*500x101 with the benchmark included

| | Demeaned returns | | | | |
|---|---|---|---|---|---|
| Date | F1 | F2 | F3 | ... | F100 |
| 2000 | -3% | -5% | -8% | | -10% |
| 2001 | -5% | 0% | -10% | | 0% |
| 2002 | -15% | -10% | -6% | | -2% |
| 2003 | 20% | 12% | 28% | | 12% |
| 2004 | 2% | 5% | 5% | | -5% |
| 2005 | -5% | -8% | -10% | | -3% |
| 2006 | 2% | 2% | 12% | | 10% |
| 2007 | 2% | -1% | 0% | | 8% |
| 2008 | -24% | -30% | -32% | | -40% |
| 2009 | 15% | 18% | 13% | | 11% |
| 2010 | 12% | 5% | 3% | | 6% |
| 2011 | 2% | 8% | 5% | | 1% |
| 2012 | 0% | 5% | -2% | | -1% |
| 2013 | 8% | 2% | 5% | | 15% |
| 2014 | 8% | 2% | 10% | | 1% |
| 2015 | -12% | -12% | -20% | | -4% |
| Average | 0% | 0% | 0% | | 0% |
| T-stat | 0.00 | 0.00 | 0.00 | | 0.00 |

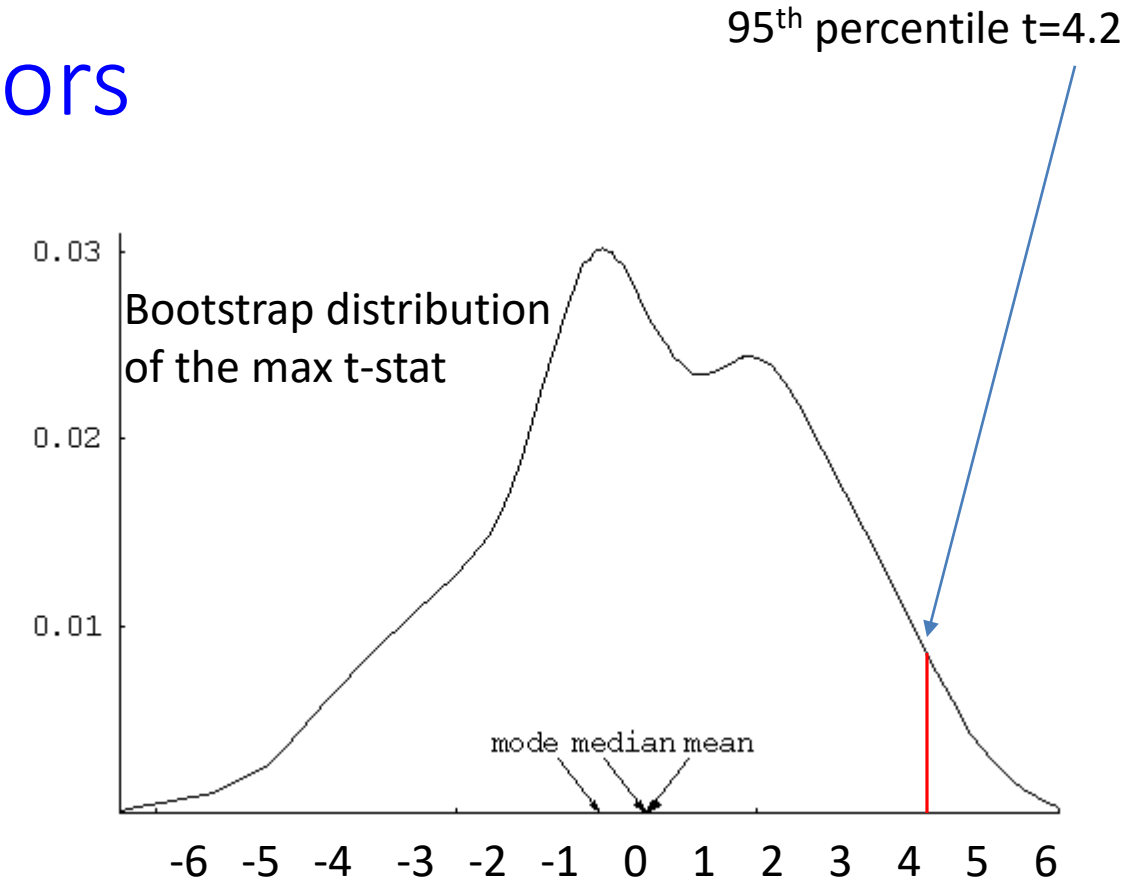
| | New history #1 | | | | |
|---|---|---|---|---|---|
| Date | F1 | F2 | F3 | ... | F100 |

# 2. Bootstrapping: Lucky Factors

- Step 3. Recalculate the alphas and t-stats on new data. Save the highest t-statistic from the 100 funds. Note, in the unbootstrapped data, every t-statistic is exactly zero.
- Step 4. Repeat steps 2 and 3 10,000 times.
- Step 5. Now that we have the empirical distribution of the max t-statistic under the null of no skill, compare to the max t-statistic in real data.

# 2. Bootstrapping: Lucky Factors

- Step 5a. If the max t-stat in the real data fails to exceed the threshold (95th percentile of the null distribution), stop (no fund has skill).
- Step 5b. If the max t-stat in the real data exceeds the threshold, declare the fund, say, $F_7$, "true"



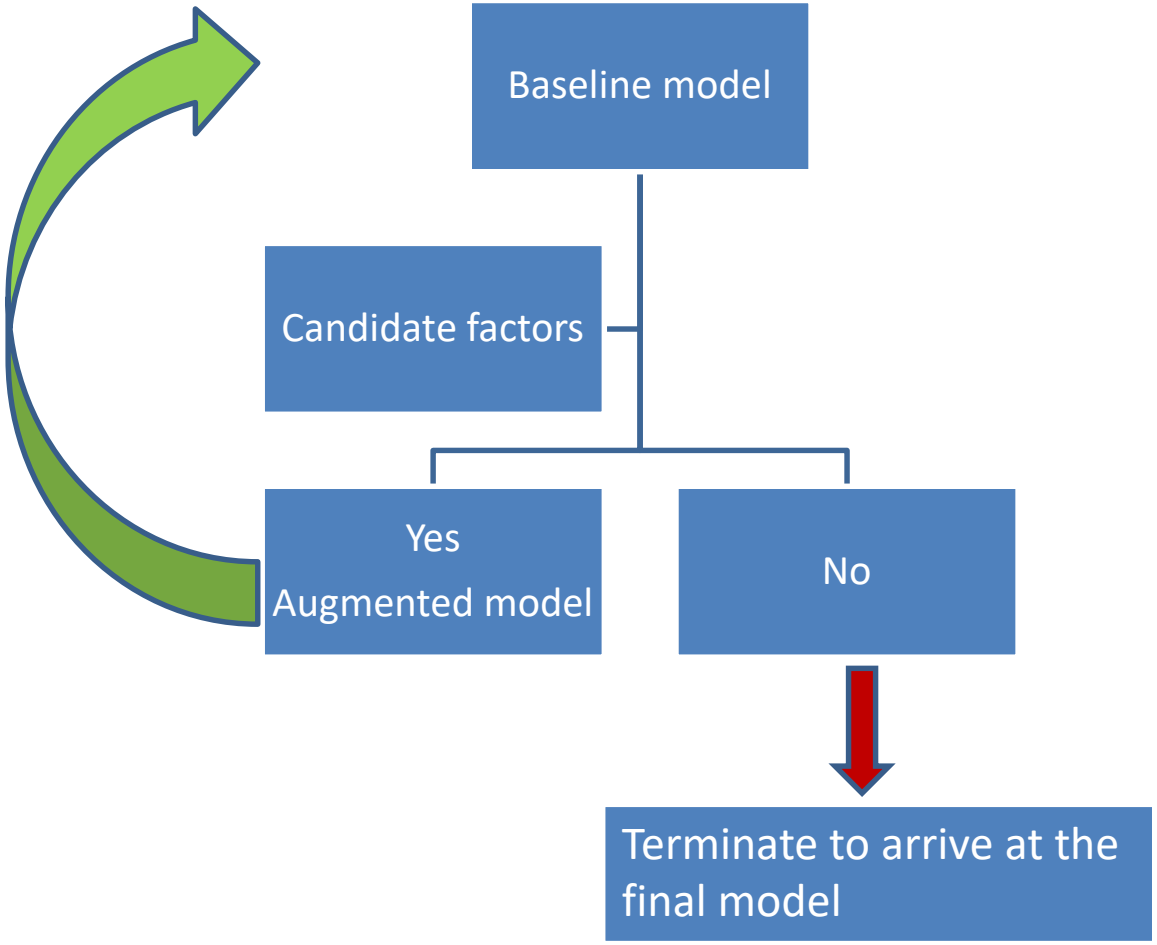
# 2. Bootstrapping: Lucky Factors

- Step 6. Replace the $F_7$ (no skill) with the actual $F_7$ (positive alpha).
- Step 7. Note that 99 funds have zero alpha and one fund has positive alpha.

# 2. Bootstrapping: Lucky Factors

- Step 8. Repeat Steps 3-5 but now we are saving the "second to max" and comparing to the second highest t-ratio in the real data.
- Step 9. Continue until the max ordered t-statistic in the data fails to exceed the max ordered from the bootstrap.

# 2. Bootstrapping: Lucky Factors



# 2. Bootstrapping: Lucky Factors

- Addresses data mining directly
- Allows for cross-correlation of the fund strategies because we are bootstrapping rows of data
- Allows for non-normality in the data (no distributional assumptions imposed – we are resampling the original data)
- Potentially allows for time-dependence in the data by changing to a block bootstrap.
- Answers the questions:
  - How many funds out-perform?
  - Which ones were just lucky?

# 2. Bootstrapping: Lucky Factors

CAMBRIDGE C|A ASSOCIATES

## Lucky Factors - A Walkthrough

*Adam Duncan*

*June 25, 2015*

The following is a walk-through of a paper called "Lucky Factors" from Campbell Harvey and Yan Liu published in 2014. It builds on a growing body of literature that is all aimed at correcting for multiple testing/comparisons that happens very frequently when researchers are building models in virtually all disciplines of science. This is particularly prevalent in finance where an incredible amount of resources have been deployed to analyze what collectively amounts to the same data set.

# 3. Noise reduction

Issue

- Past alphas do a poor job of predicting future alphas (e.g., top quartile managers are about as likely to be in top quartile next year as this year's bottom quartile managers!)
- Next lecture we will spend considerable time on implementing a noise reduction technique

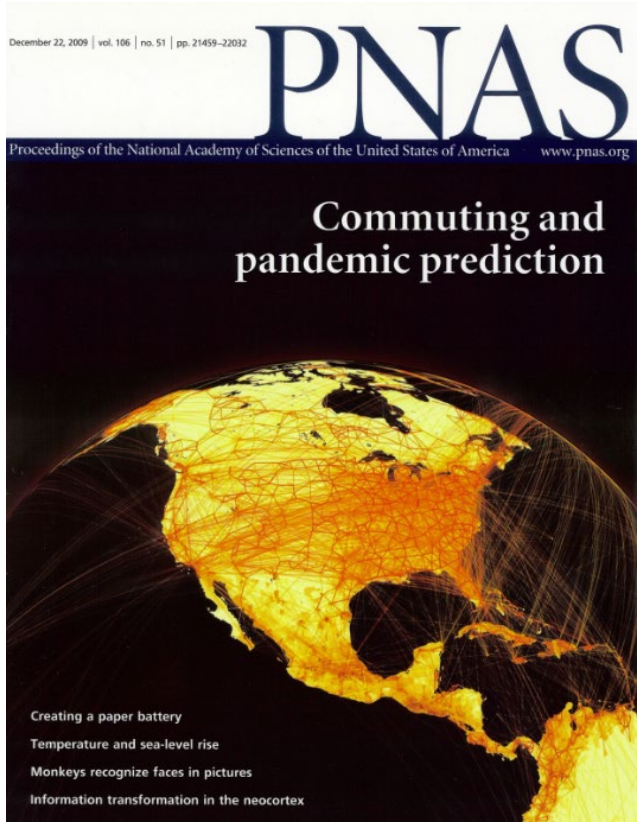
# 4. P-Hacking

- Choices are made in research that lead to a positive outcome
- This includes but goes beyond strategic data manipulation

# The Hurricane and Himicane

Jung, Shavitt, Viswanathan and Hilbe, "Female hurricanes are deadlier than male hurricanes" *Proceedings of the National Academy of Sciences*, 2014

Hypothesis: Sexism causes people take less seriously hurricanes with female names.



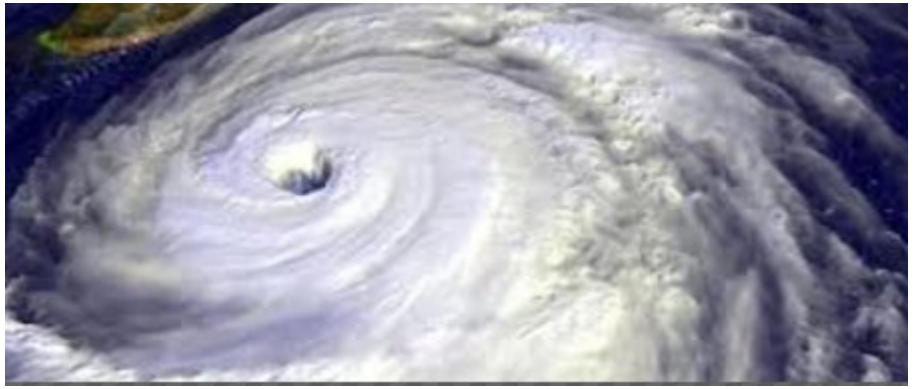

2015 Impact Factor = 9.4
Publishes 3,100 papers per year

http://www.pnas.org/content/111/24/8782

# The Hurricane and Himmicane

Jung, Shavitt, Viswanathan and Hilbe, 2014:

*"a hurricane with a relatively masculine name … is estimated to cause 15.15 deaths, whereas a hurricane with a relatively feminine name … is estimated to cause 41.84 deaths. Our model suggests that changing a severe hurricane's name from Charley … to Eloise could nearly triple its death toll."*



*"a hazardous form of implicit sexism"*

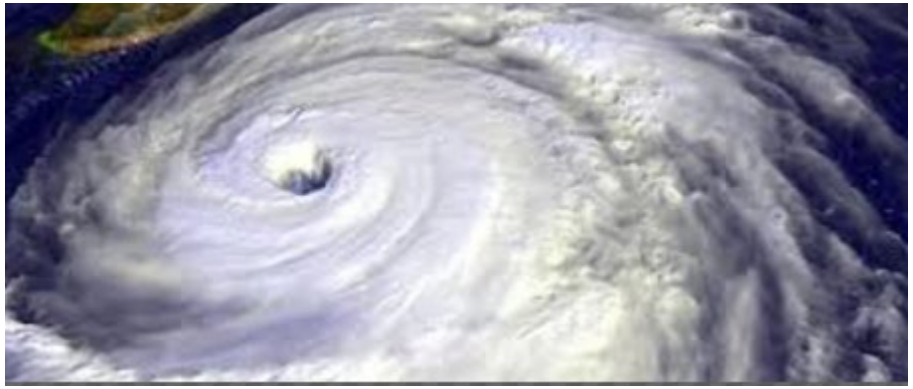
http://www.pnas.org/content/111/24/8782

# The Hurricane and Himmicane

Not Duke's Vish Viswanathan!

Jung, Shavitt, Viswanathan and Hilbe, 2014:

*"a hurricane with a relatively masculine name … is estimated to cause 15.15 deaths, whereas a hurricane with a relatively feminine name … is estimated to cause 41.84 deaths. Our model suggests that changing a severe hurricane's name from Charley … to Eloise could nearly triple its death toll."*



*"a hazardous form of implicit sexism"*

http://www.pnas.org/content/111/24/8782

# The Hurricane and Himmicane

However, certain choices were made by the researchers:

- Why exclude named tropical storms? 18 Atlantic tropical storms caused 235 deaths compared to 22 hurricanes causing 614 deaths
- Why exclude storms that do not make landfall?
- Why only count offshore fatalities if the storm comes on-shore?
- Why exclude fatalities outside the US? In 1980, Hurricane Allen made landfall near Brownsville, TX on the border of Mexico. There were 269 deaths but Jung et al. count only 2.
- Why exclude fatalities from other countries?
- Why not test robustness of results using Pacific hurricane data?

http://dx.doi.org/10.1016/j.wace.2015.11.006

# The Hurricane and Himmicane

Gary Smith, “Hurricane names: A bunch of hot air?” *Weather and Climate Extremes*, 2016, accuses authors of:

- Arbitrary exclusion of “outliers”
- Arbitrary construction of a masculinity-femininity index on a scale of 1-11. (Sandy is considered strongly feminine – more than Edith)
- Estimation of dozens of models and cherry picking the one that gives “significant” results.
- Female names dominate early data




Impact Factor = 1.4



http://dx.doi.org/10.1016/j.wace.2015.11.0

# The Hurricane and Himmicane

Gary Smith:2016, accuses authors of:

- Dropping a key variable “years elapsed since the occurrence of hurricanes”
- Misspecification of the basic model by including monetary damages as an “explanatory variable” (monetary damage cannot ‘cause’ fatalities)
- Accuses authors of “data dredging”: “If you torture the data long enough, it will confess.”
- No significant differences when sample expanded and specification corrected




Impact Factor = 1.4

http://dx.doi.org/10.1016/j.wace.2015.11.006

# 4. P-Hacking

- Sample selection
- “Outlier” exclusion
- Data transformation (scaling)
- Variable selection and combination
- Statistical test choice
- In-sample/out-of-sample

Indeed, most academic and commercial research suffers from some form of p-hacking.

# The Power Pose

Hypothesis: Standing in a posture of confidence impacts testosterone and cortisol levels in the brain leading to increased risk taking.

Evidence: Carney, Cuddy and Yap (2010)

*Psychological Science*






Carney, Cuddy and Yap, 2012. Power Posing: Brief Nonverbal Displays Affect Neuroendocrine Levels and Risk Tolerance, *Psychological Science* 21(1) 1363-1368.

# The Power Pose




Share this idea

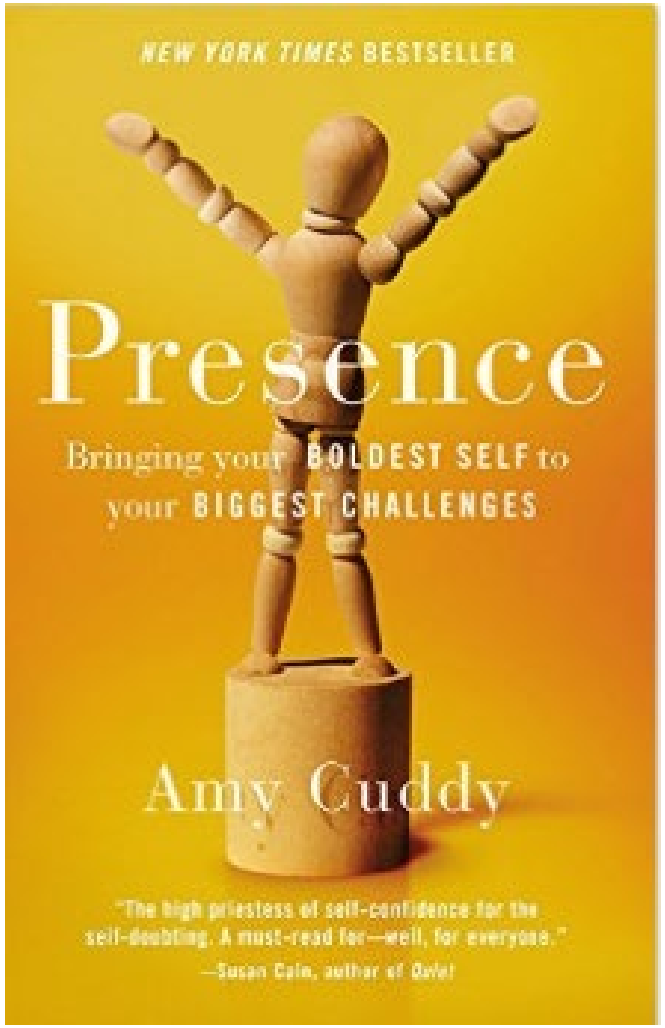
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TEDGlobal 2012 | June 2012

Second most viewed TED talk in history

# The Power Pose

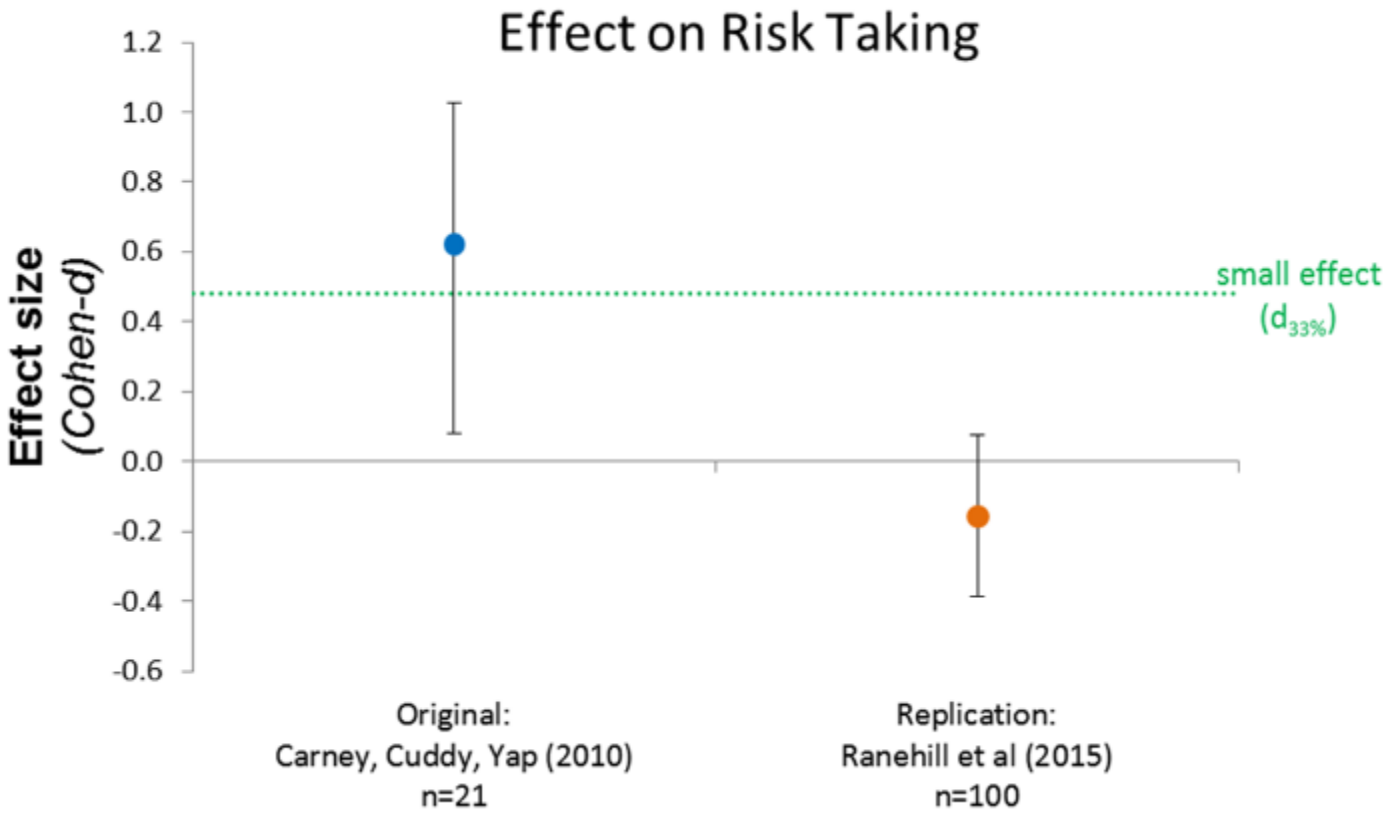



New York Times Best Seller (reached #3)

# The Power Pose

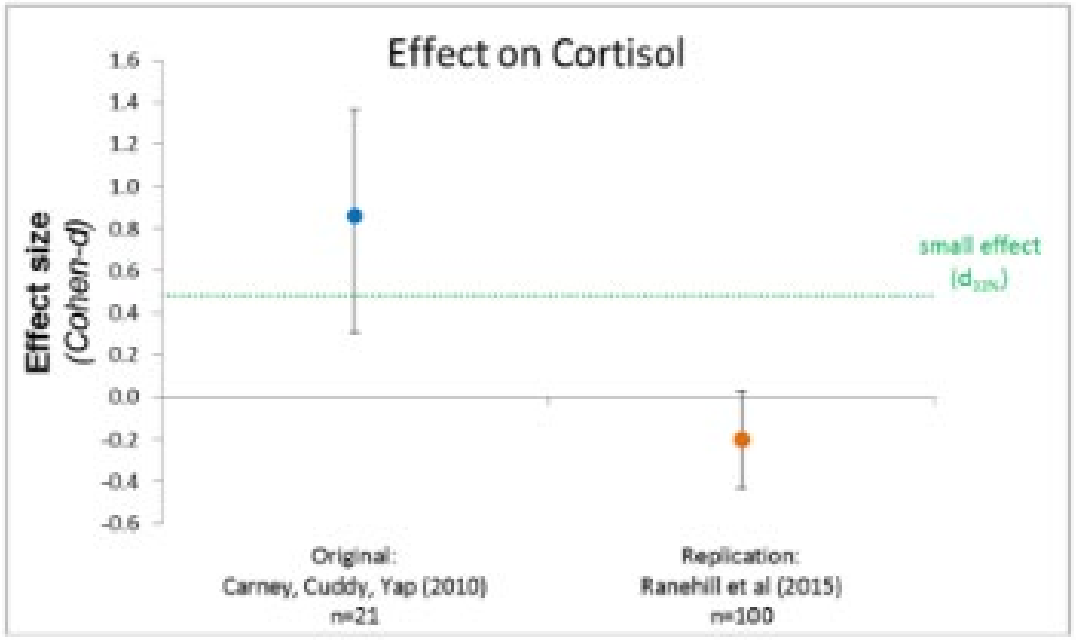


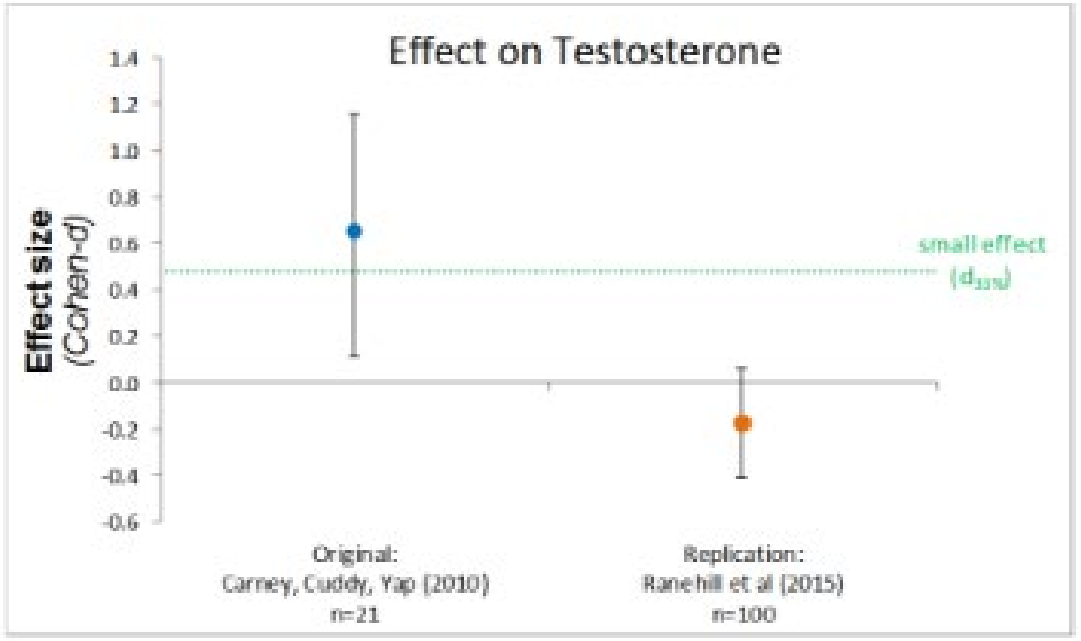
# The Power Pose




Simmons and Simonsohn, 2016, https://ssrn.com/abstract=2791272

# The Power Pose

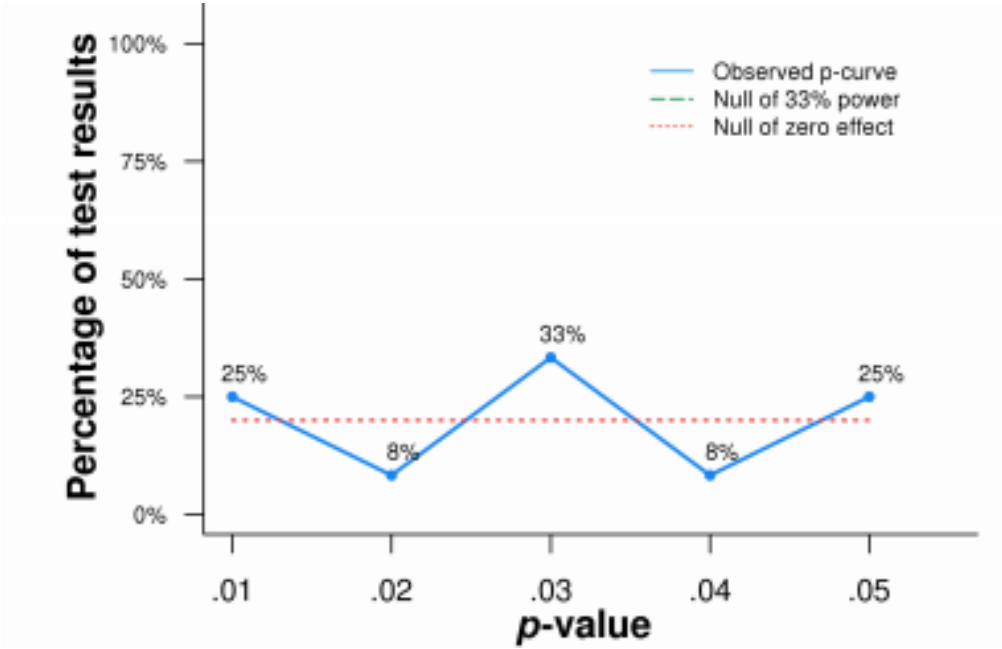



Simmons and Simonsohn, 2016, https://ssrn.com/abstract=2791272

Also see Gelman and Fung, 2016. http://www.slate.com/articles/health_and_science/science/2016/01/amy_cuddy_s_power_pose_research_is_the_latest_example_of_scientific_overreach.html

# The Power Pose




24 studies

Simmons and Simonsohn, 2016, https://ssrn.com/abstract=2791272

# The Power Pose

Dana Carney retracts

**My position on "Power Poses"**

*Regarding: Carney, Cuddy & Yap (2010).*

Reasonable people, whom I respect, may disagree. However since early 2015 the evidence has been mounting suggesting there is unlikely any embodied effect of nonverbal expansiveness (vs. contractiveness)—i.e.., "power poses" -- on internal or psychological outcomes.

As evidence has come in over these past 2+ years, my views have updated to reflect the evidence. As such, **I do not believe that "power pose" effects are real.**

http://faculty.haas.berkeley.edu/dana_carney/pdf_My%20position%20on%20power%20poses.pdf

# Rare Effects: 500 Shades of Gray

Experiment conducted at University of Virginia

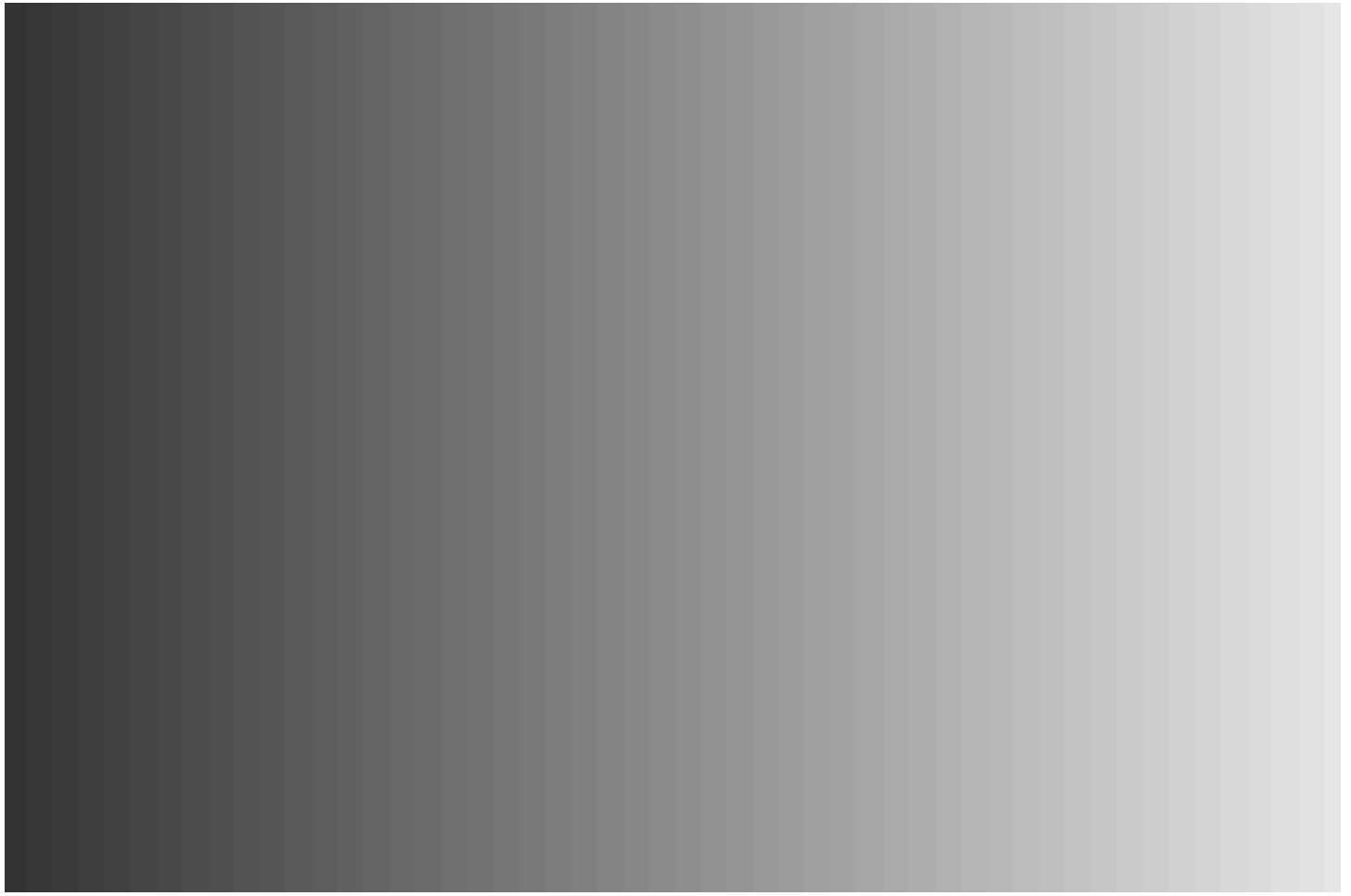
- Hypothesis: Political extremists see only black and white – literally.
- Experiment: Show words in different shades of gray and then ask participants to try to match color on gradient.
- Afterwards, evaluate where their political beliefs place on the spectrum and test hypothesis that moderates are more accurate.

Nosek, Spies and Motyl (2012)

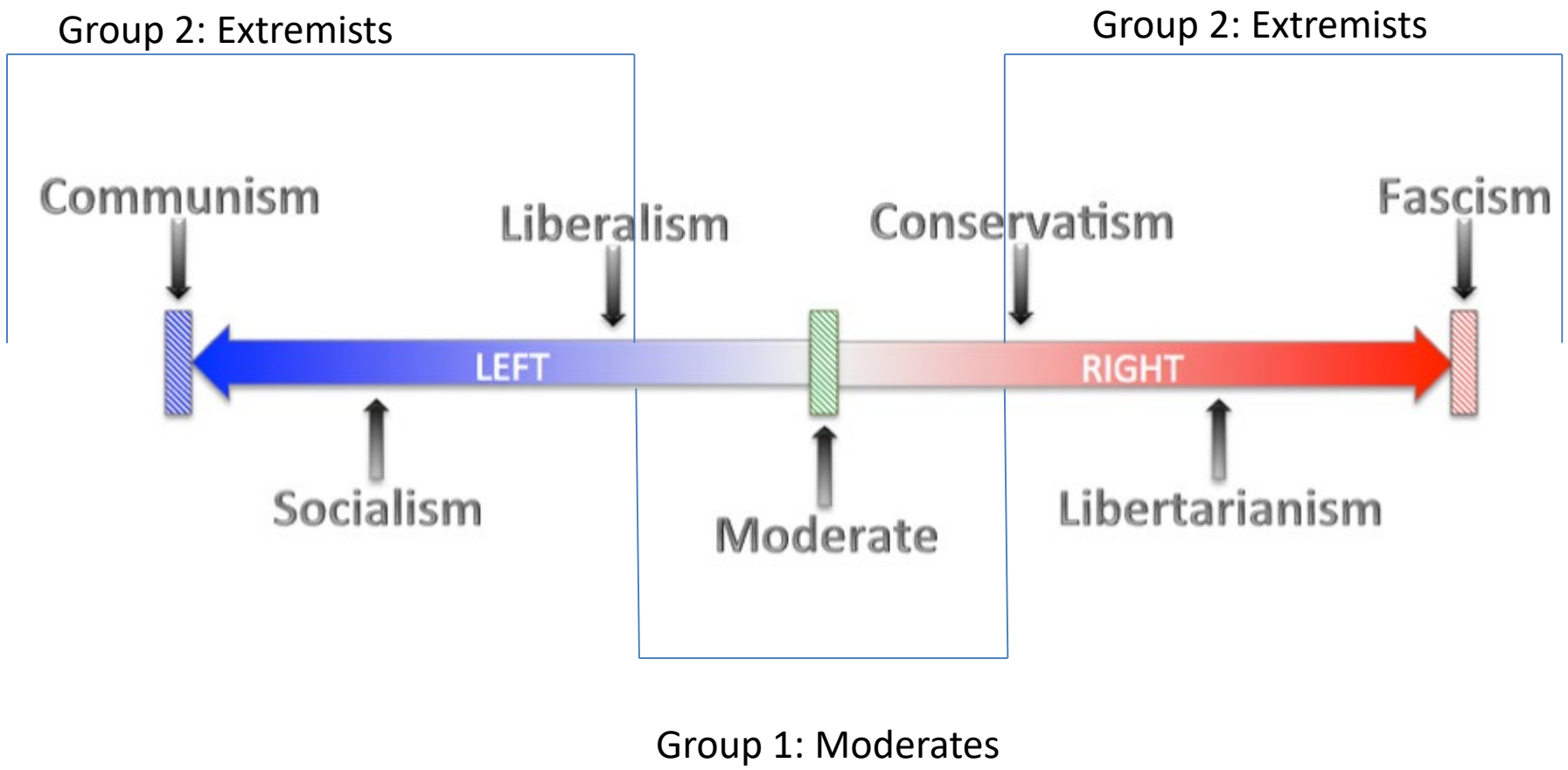
# Rare Effects: 500 Shades of Gray

Hello

Drag slide to match the color of the word



# Rare Effects: 500 Shades of Gray



# Rare Effects: 500 Shades of Gray

Dramatic results with large sample of 2,000 participants

- Moderates were able to see significantly more shades of gray
- P-value<0.001 which is highly significant; Implying only a 0.1% chance that the observed test results were consistent with the null hypothesis of no effect

# Rare Effects: 500 Shades of Gray

Researchers decided to replicate before submitting results for publication in a top journal

- Replication saw no significant difference
- P-value was 0.59  (not even close to significant)

# Rare Effects: 500 Shades of Gray

Lesson: If the hypothesis is unlikely, then we need to be especially careful. There will be a lot of false positives using standard testing procedures. Ideally, we incorporate information in the testing procedure when we know the effect is rare.

# 5. Rare Effects

- Bonferroni correction increases the threshold as a result of multiple tests
- If an effect is rare, there will be a very high Type I error rate (a lot of false positives)

# 5. Rare Effects

Go to:

**PollEv.com/finance663**

- 1% of women aged 40-50 have breast cancer – a relatively rare event
- 90% chance of a true positive test from a mammogram
- 10% error rate from mammogram

What is the chance that a woman has breast cancer given a positive test?

# 5. Rare Effects

- Survey of doctors found mean response 75%

# The Case for Using Priors

Three experiments:*
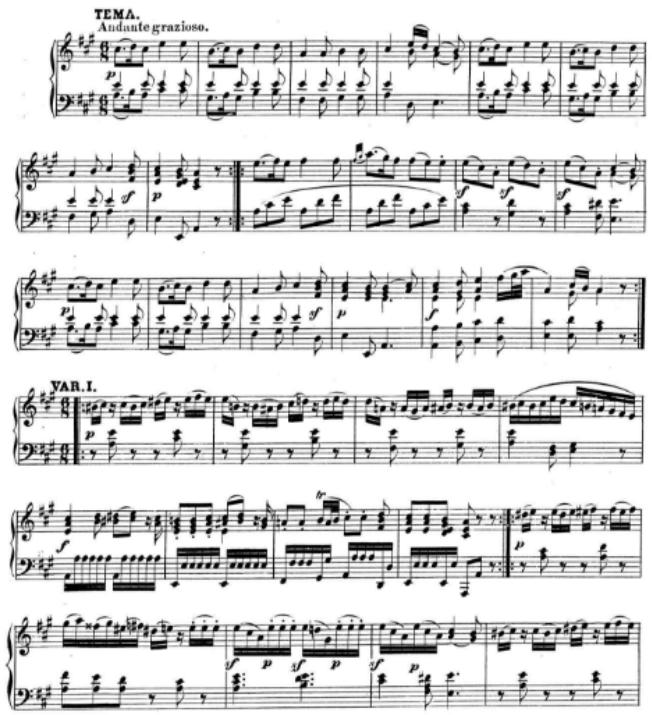
1. The musicologist
2. The tea drinker
3. The bar patron
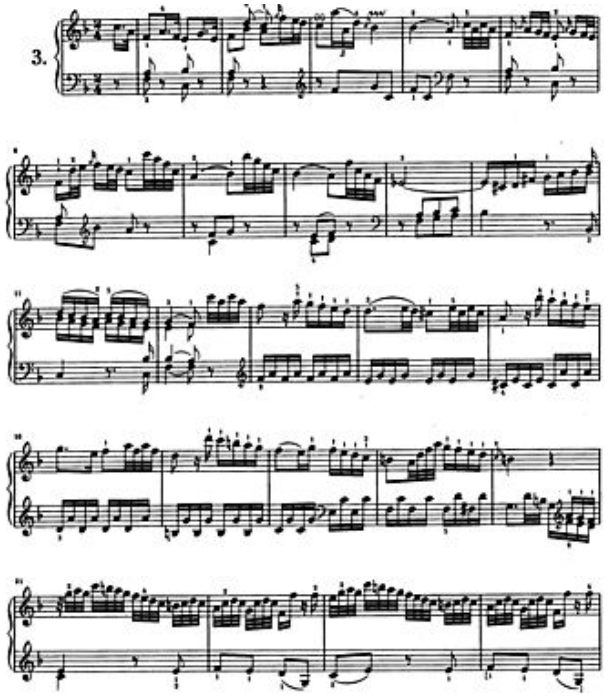


*Based on correspondence from Leonard Savage in 1962.

# The Case for Using Priors

Musicologist claims to be able to identify from unlabeled scores whether Haydn or Mozart is the composer







Simple experiment: 10 pairs of scores. Musicologist gets 10/10 correct

# The Case for Using Priors

Tea drinker claims to be able to identify whether milk was in the tea cup before the tea was poured or added afterwards





Simple experiment: 10 pairs of tea cups. The tea drinker gets 10/10 correct

# The Case for Using Priors

Bar patron claims that alcohol enables him to foresee the future



Simple experiment: Flip coin 10 times. Drunk gets 10/10 correct.

# The Case for Using Priors

All three experiments have the identical p-value:

$0.5^{10}=0.000977$ (or p-value<0.001)

- This means there is less than 1 out a 1000 chance that what we observed is consistent with the null hypothesis (no ability to choose correct answers)
- Though p-values are identical, the results have different impacts on our beliefs

# The Case for Using Priors

Three experiments:

1. The musicologist: We already know she is an expert. Indeed, it is not even clear that we need to do the experiment. Our beliefs are barely impacted.
2. The tea drinker: We might have been bit skeptical of this long time tea drinker. However, after these results, the plausibility of the claim is greatly strengthened and our beliefs shift.
3. The bar patron: The hypothesis is preposterous. P-value of 0.001 or even lower would not change our beliefs.

# Final perspectives

Combination of: propensity for Type I errors, incorrect testing methods, and lack of effort to reduce noise implies

- Most published empirical research findings are likely false
- Most research conducted within companies is likely false
- Most managers are just “lucky”
- Most the smart beta products are not “smart”
- No predictability in performance based on past performance

# Final perspectives

- My research makes progress on goal of identifying repeatable performance
- There are a host of other issues:
  - Factor loadings also noisy
  - Ex-post factor loading unfairly punish market timers
  - It is essential to look beyond the Sharpe Ratio and incorporate other info

5.